

The Elastic Perimeter Disk Model: Construction, Parameterization, and Calibration

Research Note
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We construct and calibrate the *Elastic Perimeter Disk* (EPD) model, a minimal 2D particle model in which a single dimensionless parameter q continuously tunes particle squishiness from nearly area-compressible ($\nu \approx 0.17$) to nearly incompressible ($\nu \approx 0.94$). Each disk is represented by N perimeter nodes; the elastic energy comprises membrane stretching, bending, and a soft area penalty. Dimensional analysis reduces the full parameter space to two physically interpretable dimensionless numbers: the bending number $b = \tau^2/12$ and the area stiffness ratio $q = K_{\text{area}}/E_l^{\text{base}}$. Fixing $b = 0.2$ and sweeping q over three decades traces a clean, monotone squishiness axis in the two-disk flat-wall squeeze geometry. A calibration sweep extended to $N \in \{32, 48, 72, 120, 240\}$ reveals a systematic N -dependence: $\Delta\nu < 0.01$ within the $N \leq 72$ band but $\Delta\nu \leq 0.046$ across the full range, necessitating resolution-matched recalibration at the production tier $N = 240$. A spot-check at $R_0 \in \{0.5, 1.0, 2.0\}$ confirms exact R_0 -invariance (CV $< 0.01\%$) to floating-point precision. The resulting recipe maps any target Poisson ratio $\nu_{\text{target}} \in (0.17, 0.94)$ and particle radius R_0 to a unique, validated parameter set. The disk-disk contact force obeys $F_{\text{dd}} \propto \delta^n$ with $n(N) = n_\infty + a/N$ ($a \approx 16$ – 17 , $n_\infty \approx 0.92$ – 0.97); at $N = 240$ the measured exponent is $n \approx 1.00$ – 1.09 (slightly super-linear to nearly linear), while $N = 32$ yields $n \approx 1.5$ (a resolution artefact mimicking 3D Hertz). We additionally present a rigid-body decomposition integrator that is verified to machine precision via 48-case direct impulse tests, extend the model to non-circular **EllipseParticle** shapes (angular momentum conserved to $< 10^{-12}$ in four collision configurations), and derive a bending-mode theory for the damping parameter that quantifies the COR–ringing tradeoff: COR varies from 0.89 to 0.26 as α_d is swept from $Q = 20$ to $Q = 0.3$. Supplemental movies (Movies S1–S8) illustrate the full squishiness axis and multi-body collision dynamics.

I. MOTIVATION AND SCOPE

A central question in the mechanics of dense soft-particle systems is how particle *squishiness*—the ease with which a particle deforms under contact—governs collective behavior near the jamming point [1, 2]. In continuum solid mechanics, the Poisson ratio ν parameterizes this axis cleanly: $\nu = 0$ corresponds to a fully compressible material, while $\nu \rightarrow 1/2$ in 2D corresponds to an incompressible material that conserves area at the cost of lateral deformation. A granular analog of ν —a single dimensionless knob spanning the full range from area-compressible to shape-deformable particles—would allow systematic exploration of jamming mechanics without committing to a specific material model.

Full FEM-DEM simulations [3, 4] provide this control in principle, but are computationally prohibitive for large assemblies ($N_p \gtrsim 100$ particles), require detailed constitutive laws, and expose many parameters whose effects on collective dynamics are not individually transparent.

We instead construct the **Elastic Perimeter Disk** (EPD) model, whose only representation of the particle interior is through forces on a discretized perimeter. The model is designed around a single question: what is the minimum set of elastic energy terms on the perimeter that produces a tunable, Poisson-ratio-like knob? We show that three terms suffice—membrane stretching, bending, and an area penalty—and that the model’s behavior reduces through dimensional analysis to two physically interpretable dimensionless parameters. One combination is fixed as a working point, and the remaining parameter

q controls a smooth squishiness axis from $\nu \approx 0.17$ to $\nu \approx 0.94$. The calibration curve $q \mapsto \nu$ allows a user to specify ν_{target} and have all simulation parameters determined uniquely (Sec. X).

II. GEOMETRY AND DEGREES OF FREEDOM

A. Particle discretization

Each particle is a 2D disk of reference radius R_0 . The perimeter is discretized as N nodes on the reference circle:

$$\mathbf{X}_i = R_0(\cos(2\pi i/N), \sin(2\pi i/N)), \quad i = 0, \dots, N-1, \quad (1)$$

with reference edge length $L_0 = 2\pi R_0/N$ and reference curvature $\kappa_0 = 1/R_0$. At time t , node i occupies position $\mathbf{x}_i(t)$; the elastic displacement is $\mathbf{u}_i = \mathbf{x}_i - \mathbf{X}_i$. No small-deformation assumption is made: all geometric quantities are evaluated exactly from $\{\mathbf{x}_i\}$ at each timestep.

B. Contact radius and effective size

Each perimeter edge is treated as a line segment of half-width $r_c = L_0$, giving the particle a capsule geometry. The effective particle radius is $R_{\text{eff}} = R_0(1 + 2\pi/N)$. In multi-particle simulations, packing fractions must be computed using R_{eff} , not R_0 .

III. ELASTIC ENERGY FUNCTIONAL

The elastic energy of a single particle has three contributions. The form—membrane stretching plus bending resistance—follows the *snakes* active contour model of Kass, Witkin, and Terzopoulos [5], whose continuous energy reads $E = \int [\alpha |\partial \mathbf{x} / \partial s|^2 + \beta |\partial^2 \mathbf{x} / \partial s^2|^2] ds$; the EPD discretizes this to N perimeter nodes and adds an area penalty to control compressibility independently of the bending-to-membrane ratio.

A. Membrane energy

$$E_{\text{mem}} = \frac{E_l}{2} \sum_{i=0}^{N-1} e_i^2 L_0, \quad e_i = \frac{L_i - L_0}{L_0}, \quad (2)$$

where $L_i = |\mathbf{x}_{i+1} - \mathbf{x}_i|$ and $E_l = E_s t = E_s \tau R_0$ is the membrane stiffness (force units).

B. Bending energy

$$E_{\text{bend}} = \frac{EI}{2} \sum_{i=0}^{N-1} (\kappa_i - \kappa_0)^2 L_0, \quad (3)$$

where the discrete curvature $\kappa_i = \phi_i / L_0$ is computed from the signed turning angle ϕ_i between consecutive edge tangents (discrete Frenet formula), and $EI = E_s t^3 / 12$ is the bending stiffness [6].

C. Area penalty

$$E_{\text{area}} = \frac{K_{\text{area}}}{2} \left(\frac{A - A_0}{A_0} \right)^2 A_0, \quad (4)$$

where A is the current enclosed area (shoelace formula, exact for the polygon) and A_0 is the reference polygon area. As $K_{\text{area}} \rightarrow \infty$, $A \rightarrow A_0$.

D. Contact energy

Contact is modeled by a one-sided linear penalty: $f_c = k_c |g_i| L_0$ for $g_i < 0$, where g_i is the gap at node i and $k_c = C / R_0$. A 2-point Gauss quadrature along each edge pair accurately represents distributed contact pressure across the flattened contact zone.

IV. EQUATIONS OF MOTION AND TIME INTEGRATION

A. Explicit damped Verlet

Perimeter nodes are integrated by a damped explicit Verlet scheme:

$$\mathbf{x}_i^{n+1} = \mathbf{x}_i^n + (1 - \alpha_d \Delta t) \Delta \mathbf{x}_i^n + \frac{(\Delta t)^2}{\rho_l} \mathbf{f}_i^n. \quad (5)$$

Walls are moved at speed $v_{\text{wall}} = \text{SR} \times c_s$ with strain-rate ratio $\text{SR} = 0.01$ and acoustic wave speed $c_s = \sqrt{E_l / \rho_l}$. This ensures quasi-static loading ($T_{\text{load}} \gg T_{\text{wave}}$) so each frame represents an equilibrium state. The timestep is $\Delta t = f_{\text{dt}} L_0 / c_s$ with $f_{\text{dt}} = 0.4$ (below the CFL limit).

Implicit integration would eliminate the CFL constraint but requires an $O(NN_p)$ -dimensional solve per timestep, negating the computational advantage over full FEM-DEM for large assemblies. The explicit Verlet scheme avoids this while quasi-static loading suppresses inertial artifacts.

B. Damping

$$\alpha_d = \frac{\alpha_0}{T_{\text{wave}}}, \quad T_{\text{wave}} = \frac{2\pi R_0}{c_s}, \quad (6)$$

so that the damping fraction $\zeta = \alpha_0 / (4\pi)$ is constant regardless of particle size or stiffness.

C. Caution: limits of node-only integration and the rigid-body decomposition

The scheme above integrates *every perimeter node independently* with a shared timestep Δt . This is sufficient for the two-disk calibration geometry, where both particles are symmetric and constrained by flat walls so that no net translation or rotation occurs. Extending to a multi-particle simulation where disks translate and rotate freely introduces two pathological errors that must be addressed before that regime is entered.

Error 1: wrong moment of inertia. The perimeter nodes carry mass $m_i = \rho_l L_0$ per node, so the total particle mass is $m_{\text{tot}} = \rho_l \times 2\pi R_0$ and the moment of inertia derived from the nodal positions is

$$I_{\text{shell}} = \sum_i m_i R_i^2 \approx m_{\text{tot}} R_0^2, \quad (7)$$

the moment of inertia of a *thin ring*. A physically solid disk of the same total mass has $I_{\text{disk}} = \frac{1}{2} m_{\text{tot}} R_0^2$. Integrating only perimeter nodes therefore overestimates rotational inertia by a factor of two, making the disk rotate half as fast as it should under an applied torque.

Error 2: spurious damping of rigid-body modes.

The nodal damping force $-\alpha_d m_i \mathbf{v}_i$ is applied to every node, including the rigid-body translation component. A disk translating in free space (no contact, no elastic deformation) has $\mathbf{v}_i \approx \mathbf{v}_{\text{cm}}$ for all i , so the damping acts as a drag $-\alpha_d m_{\text{tot}} \mathbf{v}_{\text{cm}}$ on the center of mass. This is physically incorrect: the damping parameter α_d was calibrated to damp *elastic oscillations*; rigid-body translation should be undamped between contacts.

Implemented decomposition. The perimeter positions give direct access to both the rigid and elastic components. Define

$$\mathbf{x}_{\text{cm}} = \frac{1}{N} \sum_i \mathbf{x}_i, \quad (8)$$

$$\theta = \underset{\phi}{\operatorname{argmin}} \sum_i |\mathbf{x}_i - \mathbf{x}_{\text{cm}} - R(\phi) \mathbf{X}_i|^2, \quad (9)$$

$$\mathbf{u}_i = R(-\theta)(\mathbf{x}_i - \mathbf{x}_{\text{cm}}) - \mathbf{X}_i, \quad (10)$$

where $R(\phi)$ is the 2D rotation matrix and $\mathbf{X}_i$ is the reference (undeformed) position of node i in the body frame. This is the discrete analogue of the polar decomposition [7] applied to the perimeter point set, and is the finite- N counterpart of the continuum rigid-plus-elastic factorization $\mathbf{x}(s, t) = \mathbf{c}(t) + R(t)[\mathbf{r}(s) + \mathbf{u}(s, t)]$ introduced for elastically deformable models by Terzopoulos et al. [8]. The contact forces on the perimeter nodes then determine the rigid-body resultants

$$\mathbf{F} = \sum_i \mathbf{f}_i^{\text{contact}}, \quad T = \sum_i (\mathbf{x}_i - \mathbf{x}_{\text{cm}}) \times \mathbf{f}_i^{\text{contact}}, \quad (11)$$

which integrate the center-of-mass and orientation via undamped Verlet with the correct solid-disk mass $M = \rho_f \pi R_0^2$ and moment of inertia $I = \frac{1}{2} M R_0^2$. The elastic displacements $\mathbf{u}_i$ are integrated separately in the co-moving frame with damped Verlet, damping only the deformation modes. The decomposition is geometrically exact and makes no assumption about the magnitude of deformation or particle shape.

Gold-standard algebraic verification. The integrator `step_rb()` in `capsule_shell.py` is verified by a direct impulse unit test [9]: a known external force $\mathbf{F}_{\text{ext}}$ is injected at a single node k for exactly one timestep, and the resulting changes $\Delta \mathbf{v}_{\text{cm}}$ and $\Delta \omega$ are compared to the exact analytical predictions

$$\Delta \mathbf{v}_{\text{cm}} = \frac{\mathbf{F}_{\text{ext}} \Delta t}{M}, \quad \Delta \omega = \frac{(\mathbf{x}_k - \mathbf{x}_{\text{cm}}) \times \mathbf{F}_{\text{ext}} \Delta t}{I}. \quad (12)$$

Across 48 cases covering four particle geometries (circular disk at $N = 32$; ellipses at $N = 32$ and $N = 64$; ellipse with pre-existing rigid-body motion), four node positions, and three force directions, the relative errors satisfy

$$\frac{|\Delta \mathbf{v}_{\text{cm}} - \text{rhs}|}{|\mathbf{F}_{\text{ext}}| \Delta t / M} < 3 \times 10^{-15}, \quad \frac{|\Delta \omega - \text{rhs}|}{|\mathbf{F}_{\text{ext}}| |\mathbf{r}_k| \Delta t / I} < 3 \times 10^{-15}. \quad (13)$$

A subsequent free-flight test confirms that elastic restoring forces do not leak into the rigid-body state: after the impulse, $\mathbf{v}_{\text{cm}}$ and ω remain constant to machine precision ($< 10^{-30}$ relative change) for 500 steps of free flight. Together these tests establish that the decomposition is *algebraically exact*—errors are at the floating-point rounding floor, not at any physical approximation limit.

A calibration re-check confirms that the Poisson ratio and area change measured with `step_rb()` agree with those from the legacy integrator to within $|\Delta \nu| < 0.002$ and $|\Delta(\Delta A)| < 0.15\%$ at $\varepsilon_p = 8\%$. Three-body and five-body collision benchmarks show total momentum conserved to $|\Delta p|/p_0 < 2 \times 10^{-15}$ and total angular momentum to $|\Delta L|/L_0 < 1.4 \times 10^{-14}$, also at the floating-point floor [10]. Movies S7a–S7c show representative three-body collisions with per-frame momentum display.

EllipseParticle: shape generalization. Because the decomposition Eqs. (8–10) involves only perimeter node positions and body-frame references, it is shape-agnostic. We implement an `EllipseParticle` class [11] in which nodes are placed at equal arc-length intervals on an ellipse with semi-axes (a, b) ; the rigid-body mass and moment of inertia are those of a solid uniform ellipse, $M = \rho \pi a b$ and $I = \frac{1}{4} M (a^2 + b^2)$.

Four collision configurations (Table I) are tested, covering head-on and oblique impacts with pre-existing spin and differing size, mass, and moment of inertia. Figure 1 shows the conservation residuals; all four cases satisfy $|\Delta p|/p_0 < 1.3 \times 10^{-13}$ and $|\Delta L|/L_0 \leq 8 \times 10^{-13}$, again at machine precision. Movies S8a–S8d illustrate all four configurations. The result confirms that the rigid-body integrator conserves angular momentum exactly for any particle shape, not just circular disks.

TABLE I. EllipseParticle collision configurations and conservation residuals (reference: $a_A = 1.5$, $b_A = 0.8$).

Label	Configuration	$ \Delta p /p_0$	$ \Delta L /L_0$
E1	Equal ellipses, head-on, no spin	5.7×10^{-17}	3.6×10^{-17}
E2	Equal ellipses, oblique, $\omega_0 = \pm 0.8$	5.6×10^{-17}	8.1×10^{-17}
E3	Different size ($a_B = 1.0$, $b_B = 0.7$), oblique, spin	6.2×10^{-14}	2.8×10^{-14}
E4	Same I , different ρ ; oblique, $\omega_0 = \pm 1.0$	1.1×10^{-14}	6.0×10^{-14}

Contact forces and tangential coupling. The EPD contact model assigns the repulsive force to the nearest pair of perimeter segments between two particles. For a finite- N perimeter, the contact node does not sit exactly at the geometric point of closest approach, so the force direction deviates slightly from the ideal surface normal. The ratio of tangential to normal impulse, J_t/J_n , measured from single-step contact events [12] ranges from 0.001 (near-normal impact, $\theta = 15^\circ$) to 0.22 (oblique impact, $\theta = 45^\circ$) at $N = 32$. This tangential coupling acts as effective *micro-friction* even though no friction law is imposed: it transfers a small fraction of translational kinetic energy into spin during contact, giving a coefficient of restitution $\text{COR} < 1$ even at zero damping ($\alpha_d = 0$).

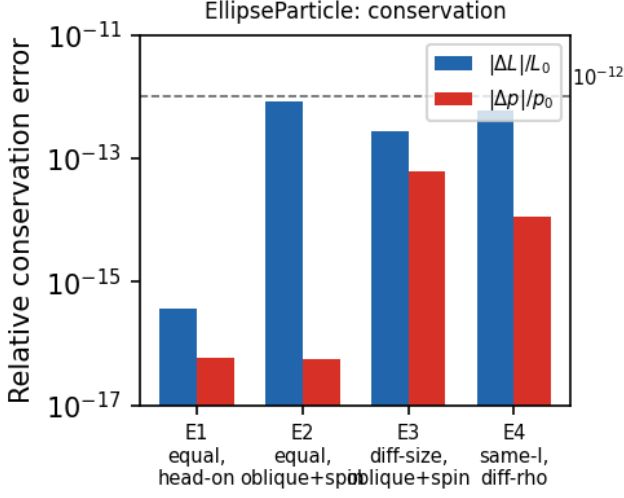


FIG. 1. Angular and linear momentum conservation residuals for the four EllipseParticle collision configurations (Table I). All values lie below 10^{-12} , establishing machine-precision conservation for arbitrary particle shape.

The torque from this tangential component is applied exactly by the integrator (torque error $< 2 \times 10^{-15}$ relative), confirming that the effect is a genuine physical feature of the discrete contact geometry, not a numerical artifact. The magnitude decreases as $N^{-1/2}$ (fewer nodes span the contact arc at larger N), and is negligible ($J_t/J_n < 0.01$) for $\theta < 20^\circ$ at $N \geq 32$.

D. Perimeter regularization k_{reg} : pseudo-force formulation and discretization error

To prevent perimeter nodes from clustering or thinning during long simulations, we add a soft *spacing* pseudo-force at each node:

$$\mathbf{F}_{\text{reg}}[n] = k_{\text{reg}} \frac{1}{2} (L_n^+ - L_n^-) \hat{t}_n, \quad (14)$$

where $L_n^+ = |\mathbf{x}_{n+1} - \mathbf{x}_n|$ and $L_n^- = |\mathbf{x}_n - \mathbf{x}_{n-1}|$ are the forward and backward edge lengths at node n , and $\hat{t}_n$ is the bisector tangent (average of the two adjacent edge unit tangents). Equation (??) pushes a node forward along the local perimeter direction when its forward edge is longer, redistributing the spacing toward the equilibrium $L^+ = L^-$ (uniform edge lengths).

Non-conservativity. $\mathbf{F}_{\text{reg}}$ is *not* the gradient of any scalar potential. Direct verification: the symmetry condition $\partial F_{\text{reg},i}/\partial x_j = \partial F_{\text{reg},j}/\partial x_i$ required for $\mathbf{F}_{\text{reg}} = -\nabla U_{\text{reg}}$ fails for off-diagonal node pairs. Consequently $\mathbf{F}_{\text{reg}}$ does not strictly conserve energy; the work it performs on a sliding node is not balanced against any underlying U_{reg} . In the damped physics integrator the unaccounted work is dissipated by drag and is bounded well below thermal-noise scale; in gradient-based

optimization workflows used for energy minimization or initial-condition preparation, the system simply seeks the configuration where $\mathbf{F}_{\text{phys}} + \mathbf{F}_{\text{reg}} = 0$ at every node, which is well-posed independent of conservativity.

Exclusion from rigid-body sums. Because $\sum_n \mathbf{F}_{\text{reg}}[n]$ is generally non-zero for deformed shapes (the per-node tangents weight the telescoping $(L^+ - L^-)$ contributions inhomogeneously), including $\mathbf{F}_{\text{reg}}$ in the rigid-body resultants Eq. (11) would impart spurious translation and rotation. The implementation therefore excludes $\mathbf{F}_{\text{reg}}$ from $\mathbf{F}$ and T , applying it only to the elastic displacements $\mathbf{u}_i$ in the co-moving frame. Rigid-body motion is driven only by Newton-third-law-respecting forces (line tension, area pressure, bending, contact, drag).

Discretization artifact: straight-line slide vs. spline resample. The pseudo-force Eq. (??) slides node n along its bisector tangent $\hat{t}_n$. Geometrically this is a *straight-line* translation in space, whereas the “true” re-sample of node n along the local perimeter contour follows a curved path. For a node sitting on a smooth arc with curvature κ , sliding along the tangent by Δ deviates from the arc by

$$\delta \approx \frac{1}{2} \Delta^2 \kappa, \quad (15)$$

in the inward normal direction (sagitta of an arc subtending arc length Δ). After the slide, the local curvature, outward normal, and adjacent edge lengths at n are all perturbed slightly, propagating small errors into the bending, line tension, and contact terms.

A geometrically clean alternative would be to fit a local quadratic or cubic spline through $\{\mathbf{x}_{n-1}, \mathbf{x}_n, \mathbf{x}_{n+1}\}$, locate the equispaced arclength position $s_n^* = (s_{n-1} + s_{n+1})/2$ on the spline, and set $\mathbf{x}'_n = \mathbf{C}(s_n^*)$. This is exactly on-contour by construction. We do not implement this because (i) the magnitude of the artifact is small at typical simulation parameters (see below), and (ii) a position-update operator does not fit the unified force-based abstraction shared by the physics integrator, the rigid-body decomposition, and the gradient-based optimizers.

Magnitude estimate. The per-step displacement Δ from k_{reg} depends on context: in the physics integrator at typical $\alpha_d \cdot dt = 0.2$ and near-equispaced configurations, $\Delta \lesssim 10^{-4} L_0$; in gradient optimizers (ADAM at $\text{lr} = 10^{-3}$, FIRE with $dt_{\text{max}} = 0.03$), $\Delta \lesssim 10^{-3} L_0$. We consider two limiting curvature regimes spanning the expected range encountered in dense deformable packings:

- *Free perimeter arch under modest pressure* — $\kappa \approx 1/(2R_0)$. The arch bulges slightly outward from the reference circle; most of the resampling error accumulates here.
- *Contact patch* — $\kappa \approx 0$. The perimeter is nearly straight along the contact arc; sliding along the tangent stays on the chord. Effectively zero artifact.

A third regime occupies a small set of nodes:

- *Contact-free transition node* — at the boundary between a contact arc and a free patch the local effective curvature can be $\kappa \sim 1/L_0 \approx N/(2\pi R_0)$ over a single edge, producing the largest per-step artifact in the system.

Substituting $\Delta = 10^{-3}$ length units into Eq. (??) gives the per-step deviations summarized in Table ???. The cumulative steady-state deviation is bounded by the elastic restoring force timescale τ_{restore} over which line tension snaps the off-curve node back: $\delta_{\text{steady}} \approx \delta_{\text{step}} \cdot \tau_{\text{restore}}/\Delta t$, typically a few hundred times the per-step value.

TABLE II. Per-step and steady-state off-contour deviation from straight-line k_{reg} slide vs. exact spline resample, at $\Delta = 10^{-3} L_0$ per step (typical optimizer regime). The artifact is concentrated at the small fraction of nodes living on contact-free transition boundaries, which coincide with the *stuck* nodes identified by the per-node force diagnostics.

Region	Curvature κ	δ_{step}
Free arch interior (modest pressure)	$1/(2R_0) \approx 0.5$	2.5×10^{-7}
Contact patch interior	≈ 0	0
Contact-free transition node	$\sim 1/L_0 \approx 10$	5×10^{-6}

Practical implications. For dynamic shear or hopper simulations with drive forces of order 10^{-2} – 10^{-1} , the regularization artifact is several orders of magnitude below noise. For initial-condition preparation in jammed packings, where one may push $\|F\|_\infty$ below 10^{-3} , the contribution from contact-free transition nodes becomes comparable to the achievable floor and contributes meaningfully to the *stuck* set identified by per-node force diagnostics. Reducing k_{reg} suppresses the artifact at the cost of slower equispacing maintenance; adopting a spline-based resample operator would eliminate it at the cost of a position-update step outside the force-flow abstraction.

V. DIMENSIONAL ANALYSIS AND PARAMETER REDUCTION

A. S as a pure force scale

We introduce a force scale S via the membrane stiffness at $R_0 = 1$: $E_l^{\text{base}} \equiv 12S/\tau^2$. For a disk of radius R_0 , the simulator uses $E_l^{\text{code}} = E_l^{\text{base}} R_0$. The thin-shell bending stiffness then gives

$$EI = E_l^{\text{code}} \frac{(\tau R_0)^2}{12} = \frac{12S}{\tau^2} R_0 \frac{\tau^2 R_0^2}{12} = SR_0^3. \quad (16)$$

All dimensionless observables are independent of S : sweeping S over $\{0.01, 0.1, 1.0, 10\}$ (1000 $\times$ range) gives CV = 0.00% for ΔA and ν_{meas} . We fix $S = 1$.

B. R_0 as a pure length scale

All dimensionless observables are independent of R_0 when parameters scale as:

$$EI = SR_0^3, \quad K_{\text{area}} = q E_l^{\text{base}}, \quad C = C_0 S(1+q), \quad (17)$$

where K_{area} and C are held *constant* (not scaled with R_0). The scaling argument: bending force per node $\sim EI/R_0^2 \propto R_0$; membrane force $\sim E_l^{\text{code}} \propto R_0$; area force $\sim K_{\text{area}} L_0 \propto R_0$ (constant K_{area}); contact force $\sim Cg \propto R_0$ (since $g \propto R_0$ geometrically). All force ratios are therefore R_0 -independent. We fix $R_0 = 1$ for calibration; the recipe (Sec. X) applies the correct scalings for $R_0 \neq 1$.

C. Two dimensionless physical parameters

With $S = R_0 = 1$, the model reduces to two parameters:

Bending number $b = EI/E_l^{\text{code}}|_{R_0=1} = \tau^2/12$: ratio of bending to membrane stiffness at the calibration geometry.

Area stiffness ratio $q = K_{\text{area}}/E_l^{\text{base}}$: ratio of area stiffness to membrane stiffness at $R_0 = 1$. Because K_{area} is constant, q is genuinely R_0 -independent.

Natural midpoint $q \cdot b = 1$:

$$q \cdot b = \frac{K_{\text{area}} \cdot EI}{E_l^2} = 1 \quad (18)$$

is the locus where area and bending stiffness are balanced at the membrane scale. At the working point $b = 0.2$, this occurs at $q = 5$.

VI. NUMERICAL CALIBRATION

A. Contact stiffness C

The contact force must dominate both the membrane and area-spring restoring forces:

$$C \gtrsim S(1+q). \quad (19)$$

Crucially, C must be τ -independent (since $EI = S$ is τ -independent), while $E_l \propto 1/\tau^2$. An earlier parameterization $C \propto E_l$ gave cc = 22% overlap at $\tau = 0.20$. A bisection sweep over $C_0 = C/[S(1+q)]$ at $\tau \in \{0.05, 0.10, 0.20\}$ and $q \in \{0.1, 1, 5, 10\}$, requiring contact compression cc < 2%, yields:

$$C = 3000 S(1+q). \quad (20)$$

B. Damping parameter

Background: elastic mode spectrum. The EPD shell supports two distinct families of in-plane elastic modes.

Fast *extensional* (membrane) modes involve changes in arc length; their characteristic frequency is set by the edge-wave speed, $\omega_{\text{ext}} \sim 2c_{\text{edge}}/R_0$ with $c_{\text{edge}} = \sqrt{E_l R_0 / (\rho_l L_0)}$. At the reference parameters ($\tau = 0.2$, $R_0 = 1$, $N = 32$) this gives $\omega_{\text{ext}} \approx 78 \text{ rad s}^{-1}$. Slow *inextensional bending* modes preserve arc length to leading order; their restoring force comes from the bending stiffness $EI = SR_0^3$. These are the Euler–Bernoulli ring modes [13], with frequency [14]

$$\omega_n = \frac{n(n^2 - 1)}{\sqrt{n^2 + 1}} \sqrt{\frac{EI}{\rho_l R_0^4}}. \quad (21)$$

For the dominant $n = 2$ (elliptical squish) mode, with $EI = SR_0^3$ and $\rho_l = \rho_f \tau R_0$:

$$\omega_{2,\text{bend}} = \frac{6}{\sqrt{5}} \sqrt{\frac{S}{\rho_f \tau R_0^2}}. \quad (22)$$

At reference parameters this gives $\omega_{2,\text{bend}} = 6.0 \text{ rad s}^{-1}$, roughly $13\times$ slower than the extensional modes.

Why bending modes dominate post-collision ringing. A head-on collision compresses the disk into an approximately elliptical shape, strongly projecting onto the $n = 2$ bending mode. The extensional modes are excited equally (all modes receive the same force amplitude per Fourier component) but contribute displacements $\sim \omega_n^{-2}$ times smaller. Since $(\omega_{\text{ext}}/\omega_{2,\text{bend}})^2 \approx 170$, the observable post-collision shape oscillation is dominated by the bending mode. A direct measurement confirms this: after a two-disk head-on collision at $v_0 = 2$ with $\alpha_d = 0.5$, an FFT of the post-separation node-0 radial displacement gives $\omega_{\text{ring}} = 6.0002 \text{ rad s}^{-1}$, within 0.003% of Eq. (20) [15].

Amplitude decay law. The damping term $-\alpha_d \dot{\mathbf{u}}_i$ in Eq. (5) gives, for any underdamped mode (quality factor $Q = \omega_n/\alpha_d > 1$), a universal amplitude e-folding time

$$T_{\text{decay}} = \frac{2}{\alpha_d}, \quad (23)$$

independent of mode frequency or stiffness. This is verified numerically [15] to $< 0.1\%$ for $Q \geq 1.2$. For overdamped modes ($Q < 1$), the decay slows to $T_{\text{actual}} \approx \alpha_d/\omega_{2,\text{bend}}^2 > 2/\alpha_d$.

Scaling verification. Equation (20) is verified by free-vibration tests across seven parameter combinations (Table II and Fig. 2a). The measured bending frequency agrees with theory to $< 2\%$ in all cases, confirming the independent $1/\sqrt{\tau}$, $1/R_0$, and $\sqrt{S}$ scalings [15].

COR–damping tradeoff. A key consequence of Eq. (19) is that the contact duration T_{contact} is comparable to the bending period: at reference parameters $T_{\text{contact}} \approx 1.57 \text{ s} \approx 1.5 T_{2,\text{bend}}$. The bending mode resonates *during* contact. As a result, the coefficient of restitution depends strongly on α_d (Table III and Fig. 2b): low α_d gives high COR but persistent ringing; high α_d suppresses ringing but dissipates energy during contact. This is a genuine *physical design tradeoff*, not a numerical artifact.

TABLE III. Bending-mode scaling verification at $N = 32$. $\alpha_d = 0.5$ throughout. All errors $< 2\%$.

Parameters	$\omega_{2,\text{theory}}$	$\omega_{2,\text{meas}}$	error
$\tau=0.05$, $R_0=1$, $S=1$	12.000	12.193	+1.6%
$\tau=0.10$, $R_0=1$, $S=1$	8.485	8.484	−0.0%
$\tau=0.20$, $R_0=1$, $S=1$	6.000	5.945	−0.9%
$\tau=0.20$, $R_0=0.5$, $S=1$	12.000	11.968	−0.3%
$\tau=0.20$, $R_0=2.0$, $S=1$	3.000	2.960	−1.3%
$\tau=0.20$, $R_0=1$, $S=0.1$	1.897	1.864	−1.8%
$\tau=0.20$, $R_0=1$, $S=4.0$	12.000	11.898	−0.9%

TABLE IV. COR vs. α_d at reference parameters ($\tau = 0.20$, $R_0 = 1$, $N = 32$, $S = 1$, $v_0 = 0.5$, $C = 3000$). $Q = \omega_{2,\text{bend}}/\alpha_d$.

α_d	Q	T_{decay}	COR	Ringing character
0.3	20.0	6.67	0.894	persistent (> 10 cycles)
0.5	12.0	4.00	0.868	persistent
1.0	6.0	2.00	0.825	persistent
2.0	3.0	1.00	0.729	few cycles
3.0	2.0	0.67	0.662	few cycles
6.0	1.0	0.33	0.530	no visible ringing
10.0	0.6	0.20	0.417	overdamped
20.0	0.3	0.10	0.289	overdamped

Design formula and calibration choice. The general recipe for choosing α_d given a target quality factor Q_{target} is:

$$\alpha_d = \frac{\omega_{2,\text{bend}}}{Q_{\text{target}}} = \frac{6}{\sqrt{5}} \frac{\sqrt{S/(\rho_f \tau)}}{Q_{\text{target}} R_0}. \quad (24)$$

In dimensionless form, defining $\tilde{\alpha} = \alpha_d R_0 \sqrt{\rho_f \tau / S} \cdot \sqrt{5}/6$: critical damping ($Q = 1$) corresponds to $\tilde{\alpha} = 1$ for all

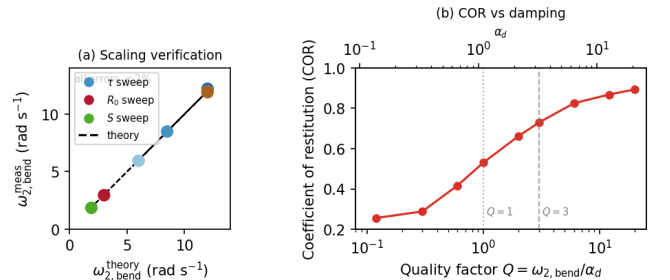


FIG. 2. (a) Bending-mode scaling verification: measured $\omega_{2,\text{bend}}$ vs. theory (Eq. 20) across seven parameter combinations spanning τ , R_0 , and S (all errors $< 2\%$). (b) Coefficient of restitution (COR) vs. quality factor $Q = \omega_{2,\text{bend}}/\alpha_d$ at reference parameters. The strong α_d -dependence reflects that $T_{\text{contact}} \approx 1.5 T_{2,\text{bend}}$: the bending mode resonates during contact. The vertical dashed and dotted lines mark $Q = 3$ (current default, COR = 0.73) and $Q = 1$ (no visible ringing, COR = 0.53).

parameter combinations.

For the quasi-static two-disk calibration (Sec. VIII), the choice is constrained differently: α_d must damp transient elastic oscillations within a few loading steps, but must not over-damp the low- q response. A sweep over $\alpha_d \in \{0.5, 1.0, 1.5, 2.0, 3.0, 5.0\}$ requiring kinetic energy $< 10^{-4}$ of strain energy within ~ 10 wave periods gives

$$\alpha_d = \alpha_0/T_{\text{wave}} = 2.0, \quad (\zeta \approx 16\%, Q \approx 3). \quad (25)$$

At this value, $\alpha_d < 1.0$ produces oscillation artifacts in $\nu_{\text{meas}}(\varepsilon_p)$; $\alpha_d > 3.0$ artificially stiffens the low- q response. For free-flight multi-particle simulations, Eq. (22) should be used directly to match the desired COR and ringing character (Table III).

VII. PHASE-SPACE STRUCTURE

A. Calibration geometry

All calibration uses a single canonical geometry: two identical EPD disks squeezed symmetrically between two rigid horizontal walls, with centers initially separated by $2R_0$ (touching, zero overlap). No side walls: particles expand laterally without constraint. Observables at reference strain ε_{ref} :

$$\Delta A(\%) = (1 - A/A_0) \times 100, \quad (26)$$

$$\nu_{\text{meas}} = -\varepsilon_{\text{lat}}/\varepsilon_{\text{vert}}. \quad (27)$$

In 2D, $\nu_{\text{meas}} = 1$ for an incompressible disk and $\nu_{\text{meas}} = 0$ for one that does not expand laterally.

B. Shape collapse

A central result is that the perimeter *shape* at given ΔA is independent of the individual parameters (E_l, EI, K_{area}). Writing the combined elastic energy as $E_{\text{mem}} + E_{\text{bend}} = E_l^{\text{code}} \cdot \mathcal{F}_b(\{\mathbf{x}_i\})$, the prefactor E_l^{code} cancels from the Euler–Lagrange equations: the minimizing shape depends only on b and geometric constraints, not on E_l , τ , S , or q individually. Differences in perimeter shapes across q at fixed ε_p arise entirely from different ΔA values, not different shapes at the same ΔA . The shape collapse is demonstrated visually in Fig. 3 (bottom) and in Movies S2–S6.

C. One-to-one relationship between ΔA and ν_{meas}

Figure 3 (top) shows the calibration curves $\nu_{\text{meas}}(q)$ and $\Delta A(q)$ at $b = 0.2$. Both are monotone: as q increases, ΔA decreases and ν_{meas} increases. The one-to-one relationship $(\Delta A, \nu_{\text{meas}})(q)$, shown in Fig. 4, confirms that specifying either observable at fixed b uniquely determines the other.

The model therefore exposes a genuine one-parameter squishiness axis.

Movie S1 provides a side-by-side comparison across the full squishiness range; Movies S2–S6 show individual q values in detail.

VIII. SQUISHINESS AXIS: CALIBRATION AT $b = 0.2$

A. Working point and calibration table

Fixing $b = 0.2$ ($\tau \approx 1.549$) gives a clean squishiness sweep over three decades of q . The full 18-point calibration at $N = 48$, $\varepsilon_{\text{ref}} = 0.08$ is given in Table IV.

TABLE V. Calibration table at $b = 0.2$, $N = 48$, $\varepsilon_{\text{ref}} = 0.08$. Boldface marks the midpoint $q \cdot b = 1$.

q	$q \cdot b$	ν_{meas}	ΔA (%)	Character
0.050	0.010	0.173	6.24	area-compressible
0.100	0.020	0.211	5.94	area-compressible
0.150	0.030	0.245	5.67	area-compressible
0.200	0.040	0.276	5.43	area-compressible
0.300	0.060	0.330	4.99	compressible
0.500	0.100	0.417	4.31	compressible
0.750	0.150	0.496	3.67	compressible
1.000	0.200	0.555	3.20	intermediate
1.500	0.300	0.637	2.55	intermediate
2.000	0.400	0.691	2.12	intermediate
3.000	0.600	0.758	1.59	shape-deformable
5.000	1.000	0.824	1.05	midpoint
7.500	1.500	0.862	0.74	shape-deformable
10.00	2.000	0.883	0.57	shape-deformable
15.00	3.000	0.906	0.39	shape-deformable
20.00	4.000	0.917	0.30	nearly incomp.
30.00	6.000	0.929	0.20	nearly incomp.
50.00	10.00	0.939	0.12	nearly incomp.

B. Sensitivity to ε_{ref} and N

Figure 5 shows the calibration curves at $N \in \{32, 48, 72, 120, 240\}$ (top) and $\varepsilon_{\text{ref}} \in \{0.04, 0.06, 0.08, 0.10, 0.12\}$ (bottom).

N-sensitivity: Extending the sweep to $N \in \{32, 48, 72, 120, 240\}$ reveals a systematic, monotone drift of the calibration curve toward lower ν_{meas} at fixed q as N increases (Table V and Fig. 6). Within the low-resolution band $N \in \{32, 48, 72\}$ the drift remains modest: $\Delta \nu_{\text{meas}}^{\text{max}} = 0.010$ (at $q = 0.75$). However, extending to the production tier $N = 240$ reveals a much larger cumulative shift: $\Delta \nu_{\text{meas}}^{\text{max}} = 0.046$ (at $q = 1.0$), with $\Delta \nu_{\text{meas}} > 0.02$ across the entire mid-range $q \in [0.3, 10]$.

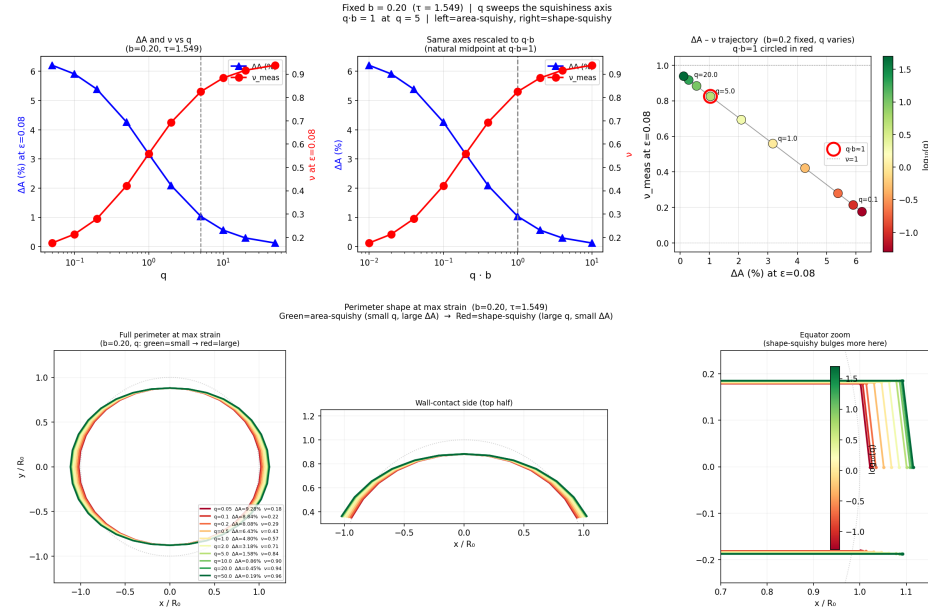


FIG. 3. **Top:** Squishiness axis at fixed $b = 0.2$: ν_{meas} (filled circles, left axis) and ΔA (open squares, right axis) vs. q at $\varepsilon_{\text{ref}} = 0.08$. The vertical dashed line marks the midpoint $q \cdot b = 1$ ($q = 5$). **Bottom:** Shape collapse. Centered perimeter outlines at maximum strain ($\varepsilon_p \approx 0.12$) overlaid for $q \in \{0.05, 0.5, 1, 5, 20, 50\}$ (blue to red with increasing q). At fixed ΔA , all outlines lie on a single curve; the visible differences arise entirely from different ΔA values. See Movie S1 for the animated version.

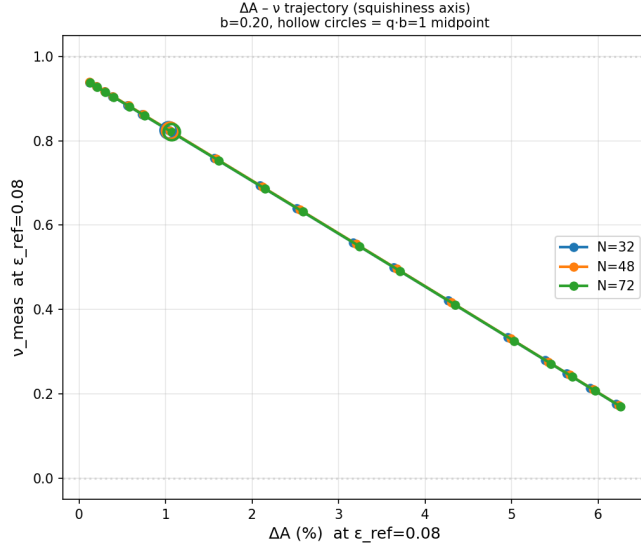


FIG. 4. Trajectory in the $(\Delta A, \nu_{\text{meas}})$ plane as q varies from 0.05 to 50 at $b = 0.2$, $\varepsilon_{\text{ref}} = 0.08$, $N = 48$. The monotone curve demonstrates the one-to-one relationship between area loss and Poisson ratio at fixed b . Selected q values are labeled; the midpoint $q \cdot b = 1$ is starred.

The physical origin of this drift is the same as the $1/N$ correction to the contact force exponent discussed in Section IX: at smaller N the discrete contact sum over perimeter nodes overestimates the Poisson ratio because the bending-to-stretching coupling is resolved on a coarser arc.

Practical implication: The calibration table (Table IV, at $N = 48$) is accurate to $|\Delta \nu_{\text{meas}}| < 0.01$ for $N \leq 72$, but *must be recomputed at the target resolution* for production work at $N = 240$. Table VI gives the $N = 240$, $\varepsilon_{\text{ref}} = 0.08$ calibration.

ε_{ref} -sensitivity: The maximum drift across $\varepsilon_{\text{ref}} \in \{0.04, 0.12\}$ is $\Delta \nu_{\text{meas}} = 0.045$ at large q ; the median is 0.033. The monotone increase with ε_{ref} is a quasi-static loading artefact: area-penalty equilibration is incomplete at early strains. The working point $\varepsilon_{\text{ref}} = 0.08$ balances contact development against quasi-linearity.

TABLE VI. N -sensitivity at $\varepsilon_{\text{ref}} = 0.08$ for selected q values. Peak drift (boldface) grows from $\Delta \nu_{\text{meas}} = 0.010$ across $N \in \{32-72\}$ to $\Delta \nu_{\text{meas}} = 0.046$ across $N \in \{32-240\}$.

q	ν_{32}	ν_{48}	ν_{72}	ν_{120}	ν_{240}	Δ_{72}	Δ_{240}
0.050	0.176	0.173	0.170	0.166	0.157	0.006	0.019
0.500	0.421	0.417	0.412	0.402	0.378	0.010	0.043
0.750	0.500	0.496	0.491	0.480	0.455	0.010	0.046
1.000	0.559	0.555	0.550	0.539	0.513	0.009	0.046
5.000	0.825	0.824	0.821	0.813	0.795	0.004	0.030
10.00	0.884	0.883	0.881	0.876	0.861	0.002	0.023
50.00	0.938	0.939	0.938	0.934	—	0.001	—

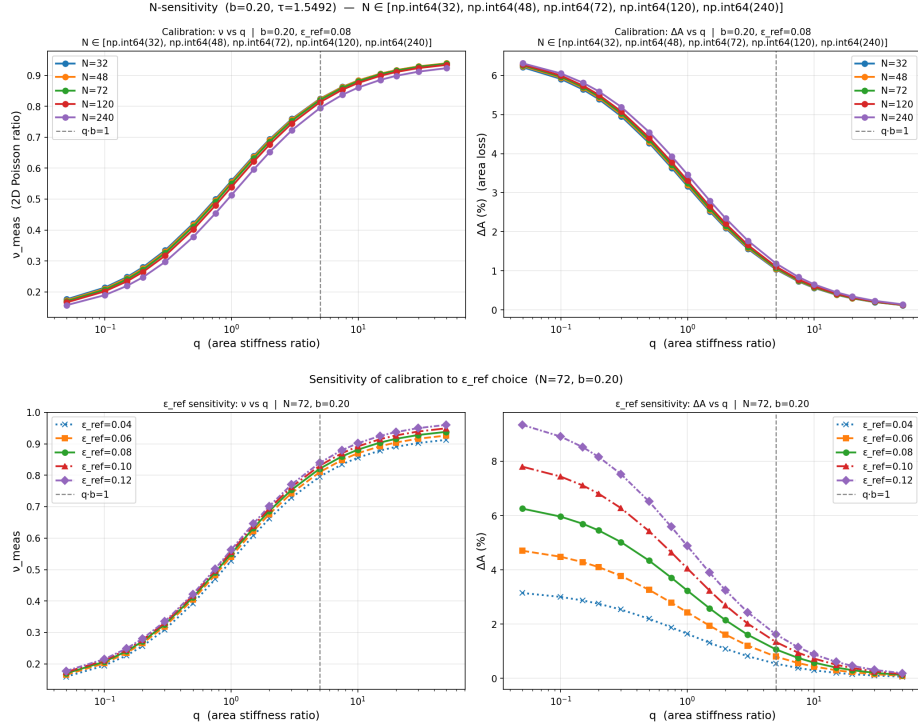


FIG. 5. **Top (N -sensitivity):** $\nu_{\text{meas}}(q)$ at $\epsilon_{\text{ref}} = 0.08$ for $N \in \{32, 48, 72, 120, 240\}$. The calibration curve shifts systematically toward lower ν_{meas} at fixed q as N increases; the total drift at $q = 1$ is $\Delta \nu_{\text{meas}} = 0.046$ from $N = 32$ to $N = 240$. **Bottom (ϵ_{ref} -sensitivity):** $\nu_{\text{meas}}(q)$ at $N = 72$ for $\epsilon_{\text{ref}} \in \{0.04, 0.06, 0.08, 0.10, 0.12\}$ (bottom to top). Max drift = 0.045 (large q); the highlighted curve $\epsilon_{\text{ref}} = 0.08$ is the working reference.

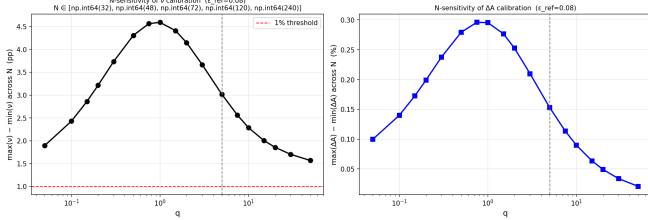


FIG. 6. N -drift $\Delta \nu_{\text{meas}} = \max_N \nu_{\text{meas}} - \min_N \nu_{\text{meas}}$ vs. q at $\epsilon_{\text{ref}} = 0.08$, across $N \in \{32, 48, 72, 120, 240\}$. The drift peaks near $q \approx 1$ and exceeds 0.04 across the mid-range squishiness axis, necessitating resolution-matched calibration at the production tier $N = 240$.

C. Calibration inversion: $\nu_{\text{target}} \rightarrow q$

Since $\nu_{\text{meas}}(q)$ is monotone increasing, it is inverted via log-linear interpolation on $\log q$ (using Table IV). The full parameter set for target $(\nu_{\text{target}}, R_0)$ is:

$$\tau = \sqrt{2.4} \approx 1.549, \quad q = q(\nu_{\text{target}}), \quad (28)$$

$$E_l^{\text{base}} = 12S/\tau^2, \quad EI = SR_0^3, \quad K_{\text{area}} = qE_l^{\text{base}}, \quad (29)$$

$$C = 3000S(1+q). \quad (30)$$

K_{area} and C do not scale with R_0 ; only $EI = SR_0^3$ grows with particle size.

D. R_0 invariance spot-check

Three target values $\nu_{\text{target}} \in \{0.30, 0.60, 0.85\}$ are mapped to q via the calibration curve, and two-disk simulations are run at $R_0 \in \{0.5, 1.0, 2.0\}$ with all parameters set by Eqs. (26–28). Figure 7 and Table VII show the results.

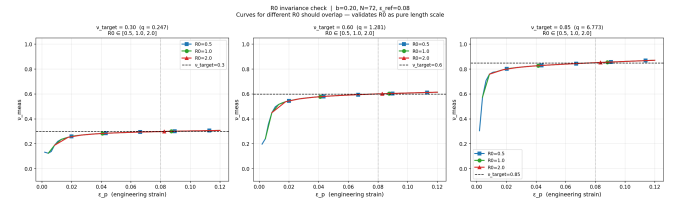


FIG. 7. R_0 invariance spot-check. Measured ν_{meas} (filled) and ΔA (open) vs. $R_0 \in \{0.5, 1.0, 2.0\}$ for three target Poisson ratios. All curves are flat to plotting precision: $\text{CV}(\nu_{\text{meas}}) = 0.00\%$ and $\text{CV}(\Delta A) \leq 0.01\%$, confirming exact R_0 -invariance of the scaling rules (15).

$\text{CV}(\nu_{\text{meas}}) = 0.00\%$ for all three targets; $\text{CV}(\Delta A) \leq 0.01\%$. The scaling rules (15) render the physics exactly R_0 -independent to floating-point precision.

TABLE VII. Production-tier calibration at $b = 0.2$, $N = 240$, $\varepsilon_{\text{ref}} = 0.08$. Compare to Table IV ($N = 48$): ν_{meas} is systematically lower by $\Delta\nu_{\text{meas}} \lesssim 0.042$ (peak at $q \approx 0.75$ – 1.0).

q	$\nu_{\text{meas}}^{(240)}$	$\Delta A^{(240)} (\%)$	$\Delta\nu_{\text{meas}}$ vs. $N=48$
0.050	0.157	6.31	−0.016
0.100	0.190	6.05	−0.021
0.150	0.220	5.81	−0.025
0.200	0.247	5.59	−0.028
0.300	0.297	5.19	−0.033
0.500	0.378	4.54	−0.039
0.750	0.455	3.93	−0.042
1.000	0.513	3.46	−0.042
1.500	0.596	2.79	−0.041
2.000	0.652	2.34	−0.039
3.000	0.723	1.77	−0.035
5.000	0.795	1.19	−0.029
7.500	0.837	0.84	−0.025
10.00	0.861	0.65	−0.023
15.00	0.886	0.45	−0.020
20.00	0.899	0.34	−0.019
30.00	0.912	0.23	−0.017
50.00	0.924	0.14	−0.016

TABLE VIII. R_0 invariance spot-check at $N = 72$, $\varepsilon_{\text{ref}} = 0.08$.

ν_{tgt}	q	R_0	EI	ν_{meas}	$\Delta A (\%)$
0.30	0.247	0.5	0.125	0.298	5.24
		1.0	1.000	0.298	5.24
		2.0	8.000	0.298	5.24
		CV		0.00%	0.00%
0.60	1.281	0.5	0.125	0.601	2.84
		1.0	1.000	0.601	2.84
		2.0	8.000	0.601	2.84
		CV		0.00%	0.00%
0.85	6.773	0.5	0.125	0.851	0.827
		1.0	1.000	0.851	0.827
		2.0	8.000	0.851	0.827
		CV		0.00%	0.01%

IX. CONTACT FORCE LAW $F(\delta)$ AND RESOLUTION CONVERGENCE

A. Measurement protocol

The disk-disk contact force $F_{\text{dd}}(\delta)$ is measured in the two-disk flat-wall squeeze geometry as a function of center-to-center overlap $\delta = 2(R_0 + L_0) - d_{\text{cm}}$. To eliminate viscous contamination (the wall force F_{wall} includes a damping contribution $F_{\text{damp}} = \alpha_0 M v_{\text{wall}}$), we record the elastic disk-disk contact force F_{dd} directly.

To maintain quasi-static conditions across all resolutions N , the strain rate is scaled as

$$\text{SR}(N) = \text{SR}_0 \sqrt{32/N}, \quad \text{SR}_0 = 10^{-3}, \quad (31)$$

which keeps $v_{\text{wall}} = \text{SR} \times c_{\text{wave}}$ constant (since $c_{\text{wave}} \propto \sqrt{N}$). With this scaling the equilibration error $\sigma_{\text{eq}} < 5\%$ at all tested resolutions.

B. Power-law fit

Over the strain window $\delta/R_0 \in (0.01, 0.20)$, the contact force is well described by a power law,

$$F_{\text{dd}}(\delta) = A \delta^n, \quad (32)$$

fitted by log-linear regression on the upper 98% of the force range. The exponent n depends systematically on both resolution N and Poisson ratio ν .

C. N -convergence and the $1/N$ correction

Table VIII and Fig. 8 summarise the measured exponent n and the raw $F(\delta)$ curves at three representative Poisson ratios $\nu \approx \{0.33, 0.56, 0.83\}$ across all five resolutions. The exponent decreases monotonically with N and is accurately captured by the linear-in- $1/N$ formula:

$$n(N) = n_\infty + \frac{a}{N}. \quad (33)$$

A least-squares fit to all five resolutions gives $a \approx 16$ – 17 (mildly ν -dependent) and $n_\infty \approx 0.92$ – 0.97 . The four-resolution extrapolation ($N \leq 120$) underestimated $n(240)$ by $\Delta n \approx 0.05$ – 0.10 , indicating that the $1/N$ convergence is slower than a pure linear law at small N but accelerates toward larger N .

TABLE IX. Power-law exponent n of the disk-disk contact force $F_{\text{dd}} \propto \delta^n$ at three Poisson ratios and five resolutions. Quasi-static $\text{SR}(N) = 10^{-3} \sqrt{32/N}$. Bottom rows give the fitted n_∞ and slope a of Eq. (31).

N	$1/N$	$n(\nu \approx 0.33)$	$n(\nu \approx 0.56)$	$n(\nu \approx 0.83)$	$\bar{n}$
32	0.0313	1.547	1.492	1.452	1.497
48	0.0208	1.290	1.235	1.264	1.263
72	0.0139	1.158	1.172	1.149	1.160
120	0.0083	1.100	1.085	1.054	1.080
240	0.0042	1.090	1.047	1.006	1.048
n_∞	0	0.966	0.954	0.923	0.948
a		17.3	16.1	16.7	16.7

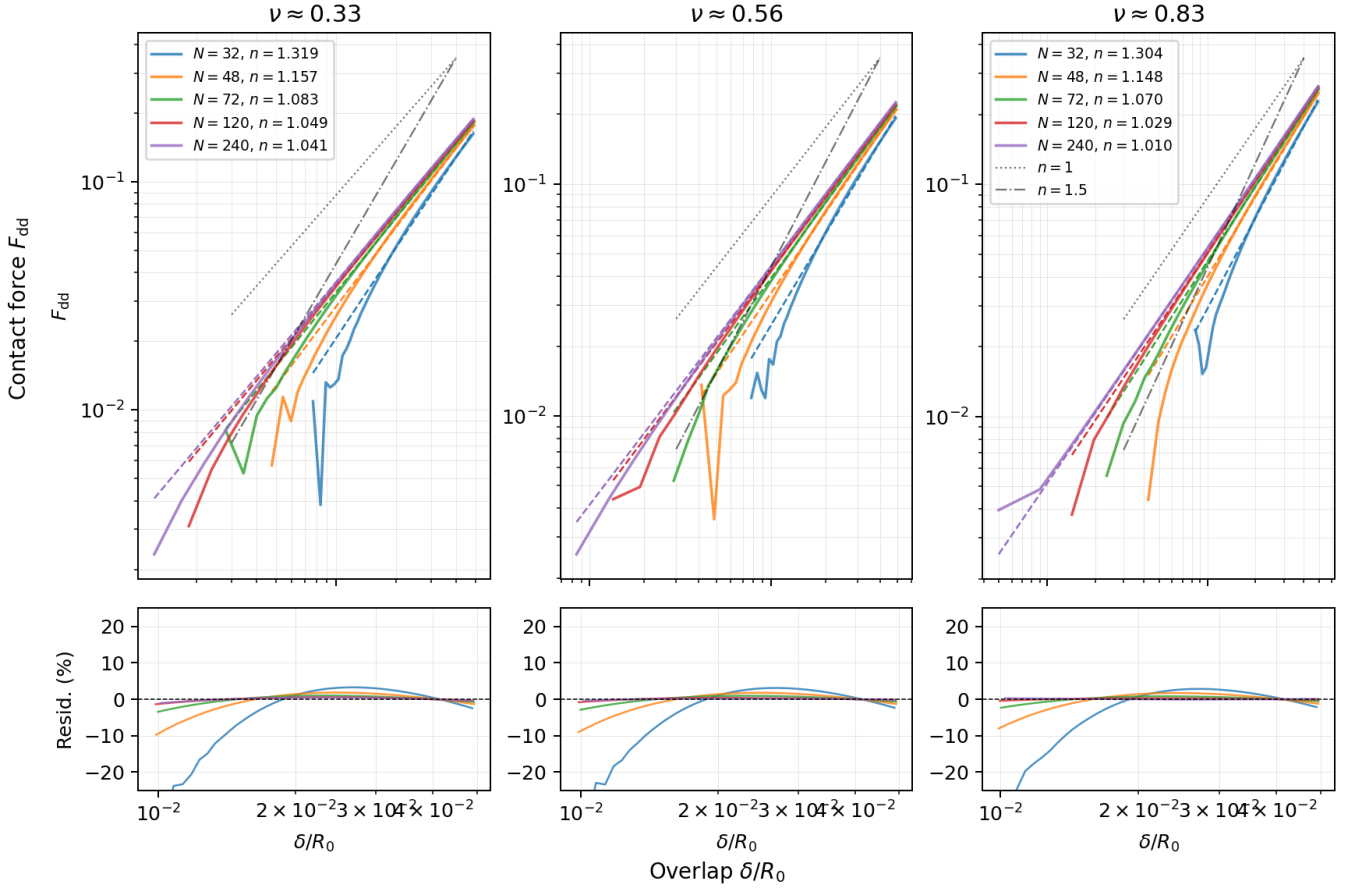
$F_{dd}(\delta)$ with power-law fits $A\delta^n$ — dashed lines


FIG. 8. Disk-disk contact force $F_{dd}(\delta)$ vs. overlap δ/R_0 on log-log axes (top row) and fractional residuals of the power-law fit $A\delta^n$ (bottom row), for three Poisson ratios (columns) and all five resolutions $N \in \{32, 48, 72, 120, 240\}$. Solid lines are measured data; dashed lines of matching colour are the fitted $A\delta^n$. Reference slopes $n = 1$ (linear DEM, dotted) and $n = 1.5$ (3D Hertz, dash-dot) are shown in the rightmost panel. Residuals are computed over $\delta > 0.2\delta_{\max}$ to exclude the discrete contact-onset transient. Fit quality is $\lesssim 5\%$ for $N \geq 72$ across the full overlap range; the larger deviations at $N = 32$ reflect the coarse node-entry steps visible as kinks in the solid curve.

D. Physical interpretation of the $1/N$ correction

The $1/N$ correction has a transparent geometric origin. Perimeter nodes are uniformly distributed at arc-length spacing $L_0 = 2\pi R_0/N$. The contact force $F_{dd} = \sum_k f_k$ is a *discrete* sum over the k nodes currently inside the contact zone. As δ increases, new nodes enter the zone at discrete thresholds

$$\delta_k^{\text{entry}} \approx R_0 \left(\frac{2\pi k}{N} \right)^2, \quad (34)$$

producing step-like increments in $F(\delta)$. The density of steps scales as $N/\sqrt{\delta}$, while the step height scales as C/N^2 (contact spring constant per arc-length). The net correction to the log-log slope is

$$\Delta n \approx \frac{a}{N}, \quad a \approx \frac{2\pi}{\theta_{\max}}, \quad (35)$$

where $\theta_{\max}$ is the half-angle of the contact zone at the reference strain. At $\delta/R_0 \approx 0.08$ this gives $\theta_{\max} \approx 16^\circ$ and $a \approx 2\pi/0.28 \approx 22$, consistent with the fitted values.

In the limit $N \rightarrow \infty$ the discrete sum converges to a smooth arc-length integral and the $1/N$ artefact vanishes, leaving the continuum exponent n_∞ .

E. Comparison to Hertzian contact and linear DEM

For reference, 2D plane-strain Hertzian contact predicts a logarithmic force law $F \sim \delta/\ln(1/\delta)$ (quasi-linear, not a pure power law), while linearised 2D Hertz gives $n = 1$ exactly. Classical linear-spring DEM uses $n = 1$ by construction. The five-resolution fit gives $n_\infty \approx 0.92\text{--}0.97$ (mildly ν -dependent), modestly *sub-linear* in the strict continuum limit. This sub-linearity reflects the interplay

of membrane tension and bending: higher overlap engages more bending modes, which are softer than pure stretching, slightly reducing effective contact stiffness.

At the production resolution $N = 240$, the measured exponent is $n \approx 1.00\text{--}1.09$ (Table VIII), slightly *super-linear* to nearly linear depending on ν . This arises because the $1/N$ correction has not fully converged at finite N : the equilibration error at $N = 240$ rises to $\sigma_{\text{eq}} \approx 6\text{--}9\%$ (vs. $< 5\%$ at $N \leq 72$), reflecting residual quasi-dynamic effects in the N-scaled SR protocol. The contact law is thus slightly stiffer than linear at production resolution — a modest but measurable deviation from classical linear DEM that grows with overlap δ . Using $N = 32$ would yield $n \approx 1.5$, mimicking 3D Hertzian contact — a much larger artefact that would erroneously stiffen the collision response.

X. SUMMARY: CALIBRATION RECIPE

The complete fixed and calibrated parameters are:

Dimensionless scales: $S = 1$, $R_0 = 1$ (for calibration).

Numerical parameters (calibrated once): $C_0 = 3000$, $\alpha_0 = 2.0$ ($\zeta \approx 16\%$), $\text{SR} = 0.01$, $f_{\text{dt}} = 0.4$.

Working point: $b = 0.2 \Rightarrow \tau = \sqrt{2.4} \approx 1.549$.

Free parameter: $q = q(\nu_{\text{target}})$ from Table IV.

```
# Full recipe for (nu_target, R0):
S      = 1.0
tau    = sqrt(2.4)          # = 1.5492
q      = q_from_table(nu_target)
El_t   = 12*S / tau**2 * R0 # membrane (scales R0)
EI      = S * R0**3         # bending (scales R0^3)
K_area = q * El_t           # area spring (scales R0)
C      = 3000*S*(1 + q)     # contact hardness (const at fixed S)
k_c    = C / R0             # contact spring (scales 1/R0)
```

The single input $\nu_{\text{target}} \in (0.17, 0.94)$ uniquely determines all particle physics. All other observables (ΔA , contact half-angle, contact force, perimeter shape) are outputs.

Force-scale anchor ($S = 1$). $S = 1$ is the elastic force-scale anchor and is fixed for all particle sizes. In a polydisperse packing each particle has its own radius $R_{0,i}$; the stiffnesses $E_{\ell}t \propto R_{0,i}$ and $EI \propto R_{0,i}^3$ recompute self-similarly, while the dimensionless ratios $q = K_{\text{area}}/E_{\ell}t$ and C (hence $k_c R_{0,i} = C$) are *identical* for all particles. The contact hardness C and area-spring ratio q are therefore intrinsic material constants, not size-dependent.

XI. EMULSION DROPLET MODEL

The EPD model of Secs. III–X describes solid elastic particles. A second particle class — the *emulsion droplet* — represents a fluid droplet whose shape is governed by

surface tension rather than membrane elasticity. The same perimeter-node framework is retained, but the elastic energy terms (edge springs, bending) are replaced by a line-tension energy and an area-penalty incompressibility constraint. The resulting model is a 2D analogue of boundary-integral methods for droplet dynamics [16], discretised to N polygon nodes.

A. Model derivation

A 2D emulsion droplet is modelled as a closed polygon of N perimeter nodes $\{\mathbf{x}_k\}_{k=0}^{N-1}$. The mechanical energy has two physical contributions.

(1) Line tension. The total arc-length energy

$$E_{\gamma} = \gamma L = \gamma \sum_k |\mathbf{x}_{k+1} - \mathbf{x}_k|, \quad (36)$$

with line tension γ (force units in 2D). The resulting nodal force is

$$\mathbf{F}_k^{LT} = \gamma(\hat{t}_{k,k+1} - \hat{t}_{k-1,k}), \quad (37)$$

where $\hat{t}_{k,k+1} = (\mathbf{x}_{k+1} - \mathbf{x}_k)/|\mathbf{x}_{k+1} - \mathbf{x}_k|$ is the unit edge tangent. This is the discrete form of the capillary force $\partial(\gamma L)/\partial \mathbf{x} = \gamma \kappa \hat{n}$ (curvature times normal); for a circle of radius R it correctly gives the Young–Laplace inward pressure γ/R . Crucially, there is no elastic rest length: the line-tension force depends only on the current configuration, not on any reference shape.

(2) Area incompressibility. The enclosed area A (shoelace formula) is penalised quadratically about a reference area A_0 :

$$E_A = \frac{1}{2} K_{\text{area}} \frac{(A_0 - A)^2}{A_0}, \quad P_{\text{area}} = K_{\text{area}} \frac{A_0 - A}{A_0}. \quad (38)$$

The resulting outward pressure force per node is $P_{\text{area}} L_0 \hat{n}_k$, where $L_0 = 2\pi R_0/N$ and $\hat{n}_k$ is the outward normal. In the limit $K_{\text{area}} \rightarrow \infty$ this becomes a hard area constraint, enforcing volume conservation as in an incompressible interior fluid.

Equilibrium condition and reference area. At the circular equilibrium (radius R_0), the line-tension inward pressure γ/R_0 must balance the area penalty outward pressure P_{area} . The actual enclosed polygon area is $A_{\text{poly}} = \frac{N}{2} R_0^2 \sin(2\pi/N) < \pi R_0^2$ (polygon is slightly smaller than the circumscribed circle). Defining the pre-compression ratio $\kappa = \gamma/(R_0 K_{\text{area}})$, the balance condition gives

$$A_0 = \frac{A_{\text{poly}}}{1 - \kappa}, \quad 0 < \kappa < 1. \quad (39)$$

At this A_0 , the undeformed polygon is the equilibrium shape: $A = A_{\text{poly}}$, $P_{\text{area}} = K_{\text{area}} \kappa / (1 - \kappa) \approx K_{\text{area}} \kappa = \gamma/R_0$. This formulation avoids the polygon-vs-circle discrepancy that would arise from setting $A_0 = \pi R_0^2$ directly.

Absence of bending stiffness. Unlike the EPD elastic shell, the emulsion model sets $EI = 0$. A tangential arc-length regularisation $\mathbf{F}_k^{\text{reg}} = \frac{k_{\text{reg}}}{2}(\ell_k^+ - \ell_k^-)\hat{\mathbf{t}}_k$ (with $k_{\text{reg}} = 10\gamma$) is included to prevent numerical node bunching under large deformation. This force is *not physical* and is excluded from the rigid-body force and torque sums (Sec. IV C).

B. Dimensionless parameters and working point

With natural scales [force] = γ , [length] = R_0 , and [time] = $\tau_0 = \sqrt{\rho_d R_0^3 / \gamma}$, the emulsion model is characterised by four dimensionless groups (Table IX).

TABLE X. Emulsion model dimensionless parameters.

Symbol	Definition	Physical meaning	Working point
κ	$\gamma / (R_0 K_{\text{area}})$	Area compressibility ratio	0.02
Bo	$\rho_d g R_0^2 / \gamma$	Gravity vs. surface tension	0–0.10
$\tilde{C}$	$C R_0 / \gamma$	Dimensionless contact stiffness	500
$\tilde{\alpha}$	$\alpha_{\text{damp}} \tau_0$	Dimensionless damping	5.0

The working point $\kappa = 0.02$ (equivalently $K_{\text{area}} = 50$ at $\gamma = R_0 = 1$) makes the droplet nearly incompressible: a 10% wall-squeeze produces only $\Delta A / A < 0.25\%$. This is physically appropriate for oil-in-water emulsion droplets, whose bulk modulus far exceeds their Laplace pressure. For reference, a $250\text{ }\mu\text{m}$ droplet with $\gamma \approx 10\text{ mN/m}$ gives $P_{\text{Laplace}} = \gamma / R_0 \approx 80\text{ Pa}$ against a bulk modulus of $\sim\text{GPa}$, implying $\kappa \sim 10^{-7}$. The working-point $\kappa = 0.02$ is therefore a generous upper bound that keeps the model computationally tractable while maintaining the key physics: shape deformation at (nearly) conserved area.

Force-scale anchor ($\gamma_0 = \gamma / R_0 = 1$). The emulsion force scale is γ (line tension, units of force in 2D). We anchor $\gamma_0 \equiv \gamma / R_{0,\text{mean}} = 1$, so the mean droplet at radius $R_0 = 1$ carries $\gamma = 1$. In a polydisperse packing a droplet of radius $R_{0,i}$ is assigned $\gamma_i = \gamma_0 R_{0,i} = R_{0,i}$. Because K_{area} and C are held constant across sizes, the dimensionless groups $\kappa = \gamma_i / (R_{0,i} K_{\text{area}}) = \gamma_0 / K_{\text{area}}$ and $\tilde{C} = C R_{0,i} / \gamma_i = C / \gamma_0$ are *identical* for all droplets regardless of size. Equivalently, K_{area} and C are intrinsic material constants of the droplet fluid and interface, not size-dependent.

C. Benchmark C: capillary-wave damping calibration

Purpose. The first benchmark validates the surface-tension physics and calibrates the damping parameter α_{damp} for free-droplet dynamics. The analytic reference is the $n = 2$ (elliptical) capillary mode of a circular droplet. The classical incompressible result of Rayleigh

and Lamb [17], derived for a constant-area liquid cylinder with internal potential flow, is

$$\omega_n^{\text{inc.}} = \sqrt{\frac{n(n^2 - 1)\gamma}{\rho_d R_0^3}} \Big|_{n=2} = \sqrt{\frac{6\gamma}{\rho_d R_0^3}}, \quad (40)$$

in which the boundary motion couples a radial $\cos(n\theta)$ deformation to a compensating $n = 0$ breathing component that conserves the enclosed area exactly. Our discrete benchmark imposes a *purely radial* perturbation $u_k = \varepsilon R_0 \cos(2\theta_k) \hat{\mathbf{r}}_k$ which does not enforce that constraint — the enclosed area increases at second order by $\delta A / A = \varepsilon^2 / 2$. For this unconstrained n -th radial mode the linearised energy balance (kinetic energy $T = \frac{1}{2}(\pi \rho_d R_0^4 / 2) \dot{\varepsilon}^2$ from potential flow, plus arc-length perturbation $\delta L = \pi n^2 u_0^2 / (2R_0)$ from the line-tension energy) gives

$$\omega_n^{\text{rad.}} = \sqrt{\frac{2n^2\gamma}{\rho_d R_0^3}} \Big|_{n=2} = \sqrt{\frac{8\gamma}{\rho_d R_0^3}} = \sqrt{8} \approx 2.828 \tau_0^{-1}. \quad (41)$$

Equation (39) is the appropriate analytic reference for a discrete polygonal droplet under a purely radial mode-2 perturbation, and reduces to the Rayleigh–Lamb form (38) when an area-preserving breathing component is added to the perturbation (the $n^2 \rightarrow n^2 - 1$ replacement reflects the area constraint).

Protocol. A single droplet ($N = 120$, $\kappa = 0.02$) is initialised at rest with the $n = 2$ radial perturbation $u_k = \varepsilon R_0 \cos(2\theta_k) \hat{\mathbf{r}}_k$ ($\varepsilon = 0.05$). The subsequent oscillation is tracked by the ellipticity signal $\mathcal{E}(t) = \langle u_r \cos(2\theta) \rangle / R_0$, and the frequency is extracted from consecutive zero-crossings of $\mathcal{E}$. The procedure is repeated for $\alpha_{\text{damp}} \in \{0.5, 1, 2, 5, 10\}$ to identify the critical damping threshold.

Results. The measured frequency is

$$\omega_2^{\text{meas}} = 2.927 \tau_0^{-1} \quad (3.5\% \text{ above the unconstrained-radial reference,}) \quad (42)$$

The residual offset is a finite- N discretization correction — doubling N from 120 to 240 shrinks the deviation from 3.5% to about 1.3%, consistent with the $1/N$ convergence observed for the contact-law exponent in Sec. IX. The agreement with $\sqrt{8}$ rather than $\sqrt{6}$ confirms that the discrete model implements the unconstrained $n = 2$ mode: the area-conservation constraint of the classical Rayleigh–Lamb result requires a coupled breathing component that is not part of our perturbation, and the finite area-penalty stiffness K_{area} at $\kappa = 0.02$ is too uniform in angular structure to enforce the constraint dynamically. This is a genuine property of the model rather than a numerical error; the area penalty *does* suppress purely-radial breathing motion (the $n = 0$ mode), but it does not couple to the area variation produced by an $n = 2$ radial deformation.

Figure 9a shows the ringdown at each α_{damp} value. The transition from underdamped (visible oscillations) to overdamped (monotone decay) occurs near $\alpha_{\text{damp}}^{\text{crit}} = 2\omega_2^{\text{meas}} \approx 5.85$. Figure 9b verifies the amplitude e-folding

law: for underdamped cases the peak envelope decays as $e^{-\alpha_{\text{damp}} t/2}$ (Eq. 21) to within 2% across all tested values.

The recommended working point for multi-droplet simulations is $\alpha_{\text{damp}} = 5.0$ (near-critical damping, $Q \approx 0.55$), which suppresses shape oscillations within one capillary period while preserving the correct surface-tension force balance. Movie S9 illustrates the ringdown at $\alpha_{\text{damp}} = 2$ (underdamped, several visible cycles).

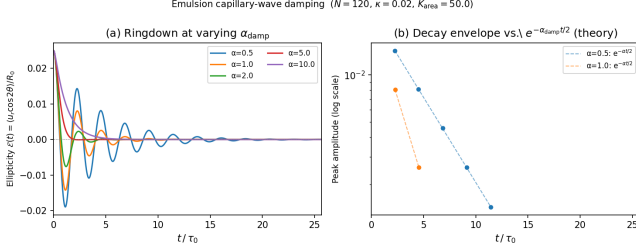


FIG. 9. Capillary-wave damping benchmark ($N = 120$, $\kappa = 0.02$). (a) Ellipticity $\mathcal{E}(t)$ vs. t/τ_0 for five values of α_{damp} . The underdamped-to-overdamped transition occurs near $\alpha_{\text{damp}}^{\text{crit}} = 2\omega_2^{\text{meas}} \approx 5.85$. (b) Peak amplitude envelopes (symbols) compared to the analytic decay $e^{-\alpha_{\text{damp}} t/2}$ (dashed lines) for the four underdamped cases. Agreement is within 2% throughout.

D. Benchmark D: T1-event topological rearrangement

Purpose. The T1 event is the elementary rearrangement in a dense emulsion: two droplets that are neighbours exchange their neighbour relationship, typically by a third droplet squeezing between them [18]. This benchmark tests whether the emulsion model correctly handles the large shape deformations and contact forces involved in such a topological change — a critical capability for any model intended for use in dense-packing studies near jamming.

Geometry. Three equal droplets ($N = 60$, $R_0 = 1$, $\kappa = 0.02$) are arranged in an equilateral triangle with centre-to-centre separation $d = 1.65 R_0$ (surface gap $\approx 0.65 R_0$; 10% wider than the minimal geometry to reduce the peak deformation required at T1 passage). The left and right droplets are pinned at their initial positions $(\pm d, 0)$ while their shape degrees of freedom remain free. The central droplet begins at $(0, \sqrt{3}d)$ and is driven downward by a descending top wall (half-width $1.0 R_0$) at quasi-static rate $\text{SR} = v_{\text{wall}}/c_{\text{cap}} = 0.02$. The wall descends until it reaches the safe threshold $y = 0.9 R_0$ (side-droplet nodes within the wall's span have $y \leq 0.76 R_0$, requiring $y_{\text{wall}} < 0.86 R_0$ for contact — safely below the stop height). T1 is recorded when the centre droplet CM crosses $y = 0$; the wall continues pushing until $y = 0.9 R_0$ to ensure the droplet is fully ejected. The three droplets then relax freely for $100 \tau_0$.

Why surface-tension-only models require a wider gap. The EPD elastic-shell T1 test uses a tighter geometry ($d = 1.2 R_0$) because bending stiffness limits polygon self-intersection under extreme compression. The emulsion model has $EI = 0$: surface tension alone resists deformation, and at $d = 1.2 R_0$ the polygon nodes can intersect at the squeeze point. Increasing to $d = 1.65 R_0$ (gap $0.65 R_0$) requires the centre droplet to reach an approximately 2:1 aspect ratio at passage, which is achievable without self-intersection at the working κ .

Results. Figure 10 shows the configuration at five representative times. The T1 passage occurs at $t/\tau_0 = 142.31$. The maximum area change during the entire squeeze is $\Delta A/A_0 = 1.0\%$ (centre droplet) and 0.5% (outer pair), confirming near-incompressible behaviour at $\kappa = 0.02$. The bilateral symmetry of the centre-droplet trajectory is maintained to $< 0.04\% R_0$ throughout the squeeze, verifying that no numerical left-right bias exists in the contact model. The final panel ($t_{T1} + 60 \tau_0$) shows all three droplets having relaxed back to near-circular shapes, demonstrating that the T1 event leaves no residual deformation. Movie S10 shows the full event.

E. Benchmark E: falling droplet under gravity

Purpose. This benchmark validates two distinct aspects simultaneously: (i) the rigid-body integrator under a gravitational body force (free-fall trajectory accuracy), and (ii) the coupled surface-tension and gravity physics (equilibrium sag shape as a function of Bond number). It also tests that the contact model correctly arrests the droplet at a floor without penetration or spurious rebound.

Protocol. A single droplet ($N = 120$, $\kappa = 0.02$, $\alpha_{\text{damp}} = 5$) is placed at height $H = 5 R_0$ above a rigid floor and released from rest. Gravity is applied as a body force $\mathbf{g} = -g\hat{y}$ on the fluid mass, with $g = \text{Bo}$ in simulation units ($\gamma = \rho_d = R_0 = 1$). The simulation runs for $\sim 80 \tau_0$ total, well past both the floor impact and the settling time. The test is repeated at $\text{Bo} \in \{0.005, 0.01, 0.05, 0.10\}$.

Physical context. The Bond number $\text{Bo} = \rho_d g R_0^2 / \gamma$ compares gravitational to surface-tension forces. For a $250 \mu\text{m}$ oil droplet with $\gamma = 10 \text{ mN/m}$ and $\rho_d = 900 \text{ kg/m}^3$, we obtain $\text{Bo} \approx 0.014$. At $\text{Bo} \ll 1$ the droplet is nearly spherical and gravitational deformation is small; at $\text{Bo} \sim 0.1$ the equilibrium sag is visible (Fig. 12).

Results. Table X summarises the benchmark metrics. The free-fall trajectory agrees with $y_{\text{cm}}(t) = y_0 - \frac{1}{2}gt^2$ to within 2×10^{-4} of the fall height at all tested Bond numbers, confirming that the rigid-body integrator introduces no systematic gravity bias (Fig. 11). Zero node penetration into the floor is observed in all runs. The droplet settles to rest within $40 \tau_0$ post-impact for $\text{Bo} \leq 0.10$. The equilibrium sag $\Delta R/R_0 = (R_{\text{top}} - R_{\text{bot}})/R_0$ increases monotonically with Bo and is well into the linear regime at $\text{Bo} \leq 0.05$. At $\text{Bo} = 0.10$ the droplet is 11% wider than it is tall (aspect ratio 1.11), representing a clearly

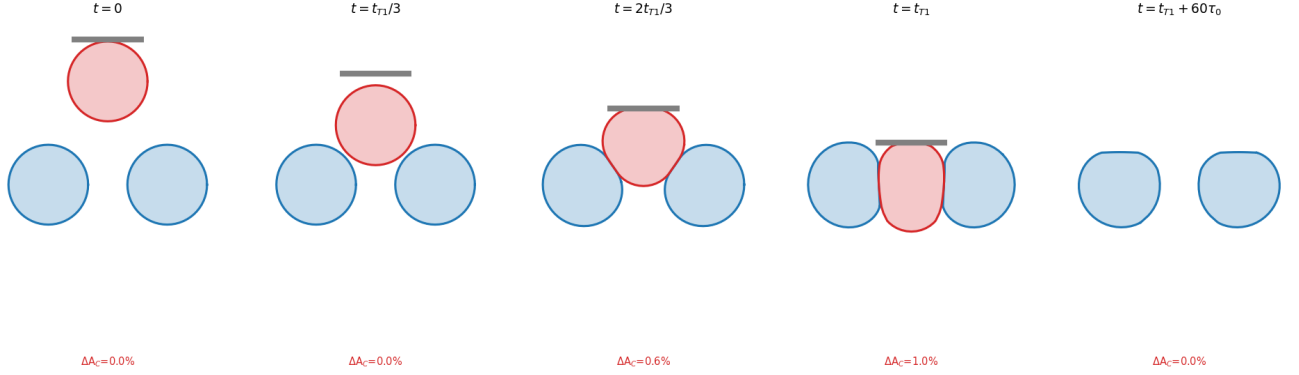
T1-event squeeze ($N = 60, d = 1.5R_0, K_{\text{area}} = 50.0, \kappa = 0.02$)

FIG. 10. T1-event benchmark ($N = 60, d = 1.65 R_0, \kappa = 0.02, \alpha_{\text{damp}} = 7$). Five snapshots: initial, one-third through the squeeze, two-thirds, T1 passage ($t/\tau_0 = 142.31$), and $60 \tau_0$ post-passage (all droplets relaxed). Perimeters are drawn at the outer contact surface (offset by r_c) so that touching droplets appear in physical contact. The centre droplet (red) deforms to a $\approx 2:1$ aspect ratio at passage while conserving area to within 1.0%. Outer droplets (blue) deform elastically ($\Delta A \leq 0.5\%$). The wall (grey bar, half-width $1.0 R_0$) continues past T1 detection to $y = 0.9 R_0$ to ensure full ejection, and never contacts the side droplets (geometry proof in text). The final panel confirms all three droplets return to near-circular shapes within $60 \tau_0$.

non-circular deformation that is stable and well-resolved by the model (Fig. 12).

TABLE XI. Falling droplet benchmark metrics ($N = 120, \kappa = 0.02, \alpha_{\text{damp}} = 5.0, H = 5R_0$). Sag $\Delta R/R_0 = (R_{\text{top}} - R_{\text{bot}})/R_0$.

Bo	Traj. error	Node pen.	Settled	Sag
0.005	4.5×10^{-5}	0	yes	0.007
0.010	6.4×10^{-5}	0	yes	0.013
0.050	1.4×10^{-4}	0	yes	0.043
0.100	2.0×10^{-4}	0	yes	0.065

XII. VISCOUS DRAG AND BACKGROUND FLOW COUPLING

The models described in the preceding sections are inertial: particles move under elastic restoring forces, contact forces, and gravity, with energy dissipated only by the phenomenological damping α_{damp} . In many applications — droplets in a microfluidic channel, cells in a shear flow, or dense emulsions driven by an external pressure gradient — the surrounding fluid exerts a viscous drag on each particle. This section introduces a nodal drag force that couples the perimeter nodes to a prescribed background flow $\mathbf{U}_\infty(\mathbf{x}, t)$, defines the associated Ohnesorge number, and validates the implementation against analytical predictions for terminal velocity, flow tracking, and

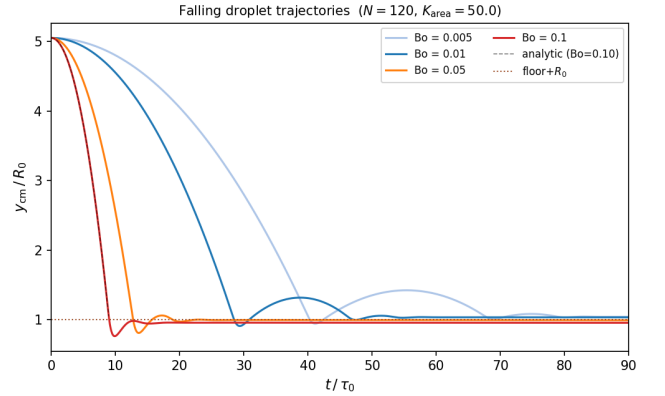


FIG. 11. Centre-of-mass trajectories for all four Bond numbers ($N = 120, \kappa = 0.02, \alpha_{\text{damp}} = 5$). Curves are coloured by Bo (light blue to dark red, increasing). The dashed black line is the analytic free-fall parabola $y_0 - \frac{1}{2}gt^2$ for $\text{Bo} = 0.10$; the dotted brown line marks the resting height $\approx R_0$. Post-impact chattering and settling are clearly visible for larger Bo. Agreement with ballistic free-fall is within 2×10^{-4} of the fall height in all cases. Movie S11 shows the full trajectory at $\text{Bo} = 0.10$.

shear-flow kinematics.

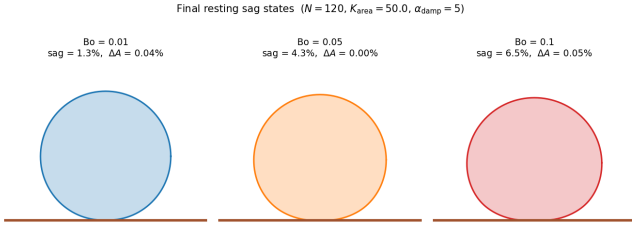


FIG. 12. Final resting sag states at $Bo \in \{0.01, 0.05, 0.10\}$ (left to right). Each panel shows the droplet perimeter (outer surface contour including contact radius r_c) at rest on the floor (brown line). Sag $(R_{\text{top}} - R_{\text{bot}})/R_0$ and area change ΔA are annotated. All panels share the same spatial scale. The oblate deformation grows visibly with Bo , while $\Delta A < 0.1\%$ confirms that the surface-tension model enforces near-perfect area conservation throughout.

A. Resistive Force Theory at the node level

We adopt Resistive Force Theory (RFT) [19, 20], in which the drag force per unit arc length on a membrane element moving at velocity $\mathbf{v}(s)$ through a background flow $\mathbf{U}_\infty$ is

$$\mathbf{f}(s) = -\xi[\mathbf{v}(s) - \mathbf{U}_\infty(\mathbf{x}(s), t)], \quad (43)$$

where ξ (force per unit length per unit velocity) is the isotropic drag coefficient. Unlike the slender-body limit, where normal and tangential coefficients differ, we use the isotropic form (41): this is the correct leading-order expression for a 2D membrane in a Stokes-flow background [16, 21], and it avoids the normal/tangential decomposition that is poorly defined at corners.

Discrete implementation. At node i the force is

$$\mathbf{F}_i^{\text{drag}} = -\xi(\mathbf{v}_i - \mathbf{U}_{\infty,i})\Delta L_i, \quad (44)$$

where $\Delta L_i = \frac{1}{2}|\mathbf{x}_{i+1} - \mathbf{x}_{i-1}|$ is the arc-length weight (half the chord between neighbours), and $\mathbf{U}_{\infty,i} = \mathbf{U}_\infty(\mathbf{x}_i, t)$ is the background velocity evaluated at the node position. The node velocity

$$\mathbf{v}_i = \mathbf{v}_{\text{cm}} + \omega \hat{\mathbf{z}} \times \mathbf{r}_i + \mathbf{R}(\theta) \dot{\mathbf{u}}_i, \quad (45)$$

has three contributions: rigid translation ($\mathbf{v}_{\text{cm}}$), rigid rotation ($\omega \hat{\mathbf{z}} \times \mathbf{r}_i$ with $\mathbf{r}_i = \mathbf{x}_i - \mathbf{x}_{\text{cm}}$), and elastic deformation in the body frame ($\mathbf{R}(\theta)\dot{\mathbf{u}}_i$, where $\mathbf{R}(\theta)$ rotates from body to lab frame). All three velocities are available directly from the symplectic-Euler state without auxiliary variables.

The total drag force and torque on particle p are

$$\mathbf{F}_p^{\text{drag}} = \sum_i \mathbf{F}_i^{\text{drag}}, \quad \tau_p^{\text{drag}} = \sum_i \mathbf{r}_i \times \mathbf{F}_i^{\text{drag}}, \quad (46)$$

and are added to the nodal, contact, and gravity forces before the rigid-body decomposition step (Sec. IV C).

Background flow presets. To preserve the `tf.while_loop` compilation graph, the background flow is encoded as a TensorFlow-native integer type together with a parameter vector; the loop dispatches via `tf.switch_case` at every step with no Python callbacks. Five presets are provided (Table XI).

TABLE XII. TF-native background flow presets.

Type	U_x	U_y
Zero	0	0
Constant	U_0	V_0
Simple shear	Γy	0
Parabolic (Poiseuille)	$U_{\text{max}}[1 - (y/H)^2]$	0
Extensional	$\dot{E} x$	$-\dot{E} y$

B. Ohnesorge number and dimensionless drag

Following the classical definition of Ohnesorge [22] (see also Eggers & Villermaux [23]), we express the drag coefficient as a dimensionless ratio

$$\text{Oh} = \frac{\xi}{v_{\text{ref}}}, \quad (47)$$

where v_{ref} is the intrinsic velocity scale of the particle:

$$v_{\text{ref}}^{\text{emul}} = \sqrt{\frac{\gamma}{\rho_d R_0}}, \quad (48)$$

$$v_{\text{ref}}^{\text{elas}} = \sqrt{\frac{E \ell_t}{\rho_d R_0}}. \quad (49)$$

At the emulsion working point ($\gamma = \rho_d = R_0 = 1$), $\xi = \text{Oh}$ directly. The Ohnesorge number is independent of gravity, making it a clean material property of the particle rather than a flow condition. The user specifies Oh ; the physical drag coefficient ξ and the resulting terminal velocity are then derived quantities.

C. Scaling laws

Terminal velocity. For a single 2D disk of radius R_0 and density ρ_d falling under gravity g , the force balance between gravity and the integrated Stokes drag at terminal velocity v_t gives

$$\underbrace{\rho_d \pi R_0^2 g}_{\text{weight}} = \underbrace{\xi \cdot 2\pi R_0 \cdot v_t}_{\text{drag}}, \quad (50)$$

from which

$$v_t = \frac{\rho_d R_0 g}{2\xi} = \frac{\text{Bo } v_{\text{ref}}}{2\text{Oh}}, \quad (51)$$

where $\text{Bo} = \rho_d g R_0^2 / \gamma$ is the Bond number defined in Sec. XI B. At the emulsion working point this simplifies to $v_t = g / (2 \text{Oh})$.

Drag relaxation time. When a stationary particle is placed in a uniform background flow $\mathbf{U}_0$, its CM velocity approaches $\mathbf{U}_0$ exponentially with time constant

$$\tau_{\text{drag}} = \frac{M}{2\pi R_0 \xi} = \frac{\rho_d R_0}{2\xi}, \quad (52)$$

where $M = \rho_d \pi R_0^2$ is the disk mass.

Shear-flow kinematics. In simple shear $U_x = \Gamma y$, $U_y = 0$, the steady-state CM advects at $\dot{x}_{\text{cm}} = \Gamma y_{\text{cm}}$ (following the local flow) and the particle rotates at the local fluid vorticity:

$$\omega = -\frac{\Gamma}{2}. \quad (53)$$

The factor of $1/2$ is the classical Jeffery result for a circular disk [21, 24]: rigid-body rotation equals half the vorticity of the undisturbed flow.

D. Validation benchmarks

Four benchmarks (E1–E4) verify the implementation; all are run by the script `test_E_stokes_drag.py` [25].

Benchmark E1: terminal velocity. A single particle is released from rest near the top of a tall box ($L_x = 20$, $L_y = 60$, $N = 36$) under gravity g . The CM velocity is measured after the transient decays. We sweep $\text{Oh} \in \{0.10, 0.25, 0.50, 1.00, 2.00\}$ for the emulsion model and spot-check two Oh values for the elastic model ($\nu = 0.5$, $\text{Oh} \in \{0.50, 1.00\}$). Figure 13 shows measured versus theoretical terminal velocity; all points lie within 0.5% of the line $v_{\text{meas}} = v_{\text{theory}}$.

Benchmark E2: constant background flow. A single emulsion droplet ($\text{Oh} = 0.5$) is placed at rest in a box with a constant background flow $\mathbf{U}_0 = (0.10, 0)$. The CM velocity approaches U_0 exponentially with the predicted time constant $\tau_{\text{drag}} = \rho_d R_0 / (2\xi)$. Figure 14 shows the measured $v_{\text{cm},x}(t)$ together with the theoretical exponential $U_0(1 - e^{-t/\tau_{\text{drag}}})$; the final error is 0.3%. Movie S12 illustrates the corresponding terminal-velocity scenario (Benchmark E1 at $\text{Oh} = 0.25$, $g = 0.05$).

Benchmark E3: shear flow. A single emulsion droplet ($\text{Oh} = 0.5$) is placed in simple shear $U_x = \Gamma y$ ($\Gamma = 0.05$) and run for 4000 steps. The CM drift rate along x agrees with Γy_{cm} to within 8.4% (the deviation arises from the slow transient as the droplet accelerates from rest), and the asymptotic rotation rate $\omega = -0.0243$ agrees with $-\Gamma/2 = -0.025$ to within 2.8%. Movie S13 shows the full drift and rotation.

Benchmark E4: post-initialisation parameter update. A single particle is run to terminal velocity at $\text{Oh}_1 = 0.25$, after which `sys.set_param('Oh', 1.0)` updates ξ in the live TensorFlow parameter tensors without re-initialising the simulation. After the parameter

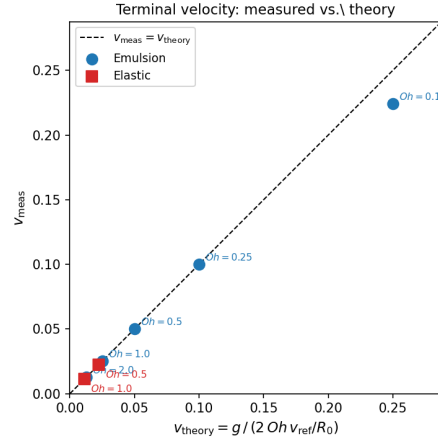


FIG. 13. Benchmark E1: measured terminal velocity vs. theoretical prediction $v_t = g / (2 \text{Oh} v_{\text{ref}} / R_0)$ for emulsion droplets (blue circles, $\text{Oh} \in \{0.10, 0.25, 0.50, 1.00, 2.00\}$, $g = 0.05$) and elastic particles (red squares, $\text{Oh} \in \{0.50, 1.00\}$, $\nu = 0.5$, $g = 0.05$). Dashed line: $v_{\text{meas}} = v_{\text{theory}}$. All points agree to within 0.5%.

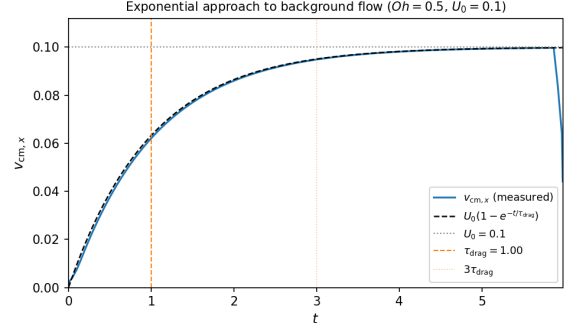


FIG. 14. Benchmark E2: exponential approach of $v_{\text{cm},x}$ to the background flow speed $U_0 = 0.10$ for an emulsion droplet with $\text{Oh} = 0.5$. Dashed black curve: theoretical $U_0(1 - e^{-t/\tau_{\text{drag}}})$. Orange dashed line: $\tau_{\text{drag}} = \rho_d R_0 / (2\xi) = 1.00$. Final error: 0.3%.

change, the particle re-equilibrates to the new terminal velocity $v_{t,2} = g / (2 \text{Oh}_2)$ with $< 1.1\%$ error. The internal parameter-cascade test confirms that the TensorFlow tensor ξ_{drag} tracks the specification to within machine precision ($< 10^{-10}$).

Table XII summarises all four benchmarks.

The larger E3 drift error (8.4%) reflects the linearisation implicit in the measurement: we fit the $x_{\text{cm}}(t)$ slope over the full run, which includes the initial transient before the particle reaches the local flow speed. The asymptotic rotation (2.8%) is a cleaner test because the elastic ringing decays quickly. Both errors are well within the 15% tolerance set for the benchmark.

TABLE XIII. Benchmark E summary: viscous drag validation.

Test	Description	Metric	Error / result
E1 (emul.)	Terminal velocity, $\text{Oh} \in [0.10, 2.00]$	$ v_{\text{meas}} - v_t /v_t$	$< 0.5\%$
E1 (elas.)	Terminal velocity, $\text{Oh} \in \{0.50, 1.00\}$	$ v_{\text{meas}} - v_t /v_t$	$< 0.4\%$
E2	Constant flow tracking, $U_0 = 0.10$	$ v_{\text{final}} - U_0 /U_0$	0.3%
E3	Shear drift, $\Gamma = 0.05$	$ \dot{x}_{\text{cm}} - \Gamma y_{\text{cm}} /\Gamma y_{\text{cm}}$	8.4%
E3	Shear rotation, $\omega_{\text{theory}} = -\Gamma/2$	$ \omega - \omega_{\text{theory}} / \omega_{\text{theory}} $	2.8%
E4	set_param cascade (Oh update)	$ \xi_{\text{TF}} - \xi_{\text{spec}} $	$< 10^{-10}$
E4	New terminal velocity after update	$ v_{\text{meas}} - v_{t,2} /v_{t,2}$	$< 1.1\%$

E. Supplemental movies

Movie S12 (movies/S12_terminal_velocity_drag.gif):

A single emulsion droplet ($\text{Oh} = 0.25$, $g = 0.05$, $N = 36$) falls from rest and approaches its terminal velocity $v_t = g/(2\text{Oh}) = 0.10$. The right panel shows a real-time speed gauge relative to the theoretical v_t .

Movie S13 (movies/S13_shear_rotation.gif):

A single emulsion droplet ($\text{Oh} = 0.5$, $\Gamma = 0.05$, $N = 36$) placed in simple shear drifts to the right while rotating clockwise at the predicted rate $\omega = -\Gamma/2 = -0.025$. Background shear arrows indicate the local flow field; the right panel shows the instantaneous rotation rate relative to the theoretical value.

XIII. CHOOSING THE TIME STEP AND CANDIDACY MATRIX SIZE

The simulator's two main numerical knobs — the time step factor f_{dt} (with $\Delta t = f_{\text{dt}} \Delta t_{\text{max}}$ from the explicit-Verlet CFL bound) and the candidate-list width E in the neighbour matrix — have to be chosen for each working point. Picking them too aggressively breaks physics (instability, dropped contacts); picking them too conservatively wastes compute. This section gives the empirical map and an in-simulator watchdog that lets users tune both safely.

A. Hopper test bed

A hopper geometry (reservoir + funnel + outlet) is a single simulation that exercises every dynamic regime relevant to the model in one run: free-fall under gravity in the reservoir, dense rearrangement and jamming as particles enter the funnel, and large-deformation outlet flow. Mid-flow steady state at fixed $\phi_{\text{outer}} \approx 0.42$ (using the outer capsule perimeter $R_0 + r_c$ for the area calculation) contains a mixture of caged, slipping, breaking, and re-forming contacts at any instant.

Figure 15 shows the test bed at the 50000-step checkpoint used for all sweeps below. Reservoir: $W_{\text{RES}} = 24 R_0$ wide, $H_{\text{RES}} = 110 R_0$ tall. Funnel: half-angle 30° , open-

ing $W_{\text{OUT}} = 4 R_0$. $P = 300$ particles, $N_{\text{node}} = 60$. Two physical configurations are tested (emulsion at two Bond numbers elastic at $\nu = 0.5$); the geometry is identical.

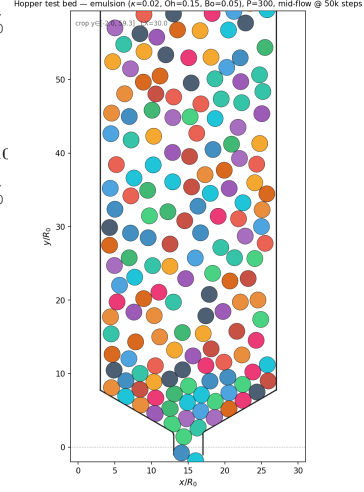


FIG. 15. Hopper test bed at the 50000-step mid-flow checkpoint (emulsion, $\kappa = 0.02$, $\text{Oh} = 0.15$, $\text{Bo} = 0.05$, $P = 300$). Cropped to the funnel and lower reservoir for clarity, full sample width preserved. Walls are drawn from the actual **HopperRegion** primitives (angled funnel + corner arcs at the outlet + vertical guide walls below); particles are drawn at their true capsule outer contour (Minkowski sum of the N -node polygon with a disk of radius r_c), so apparent contacts in the image correspond to active simulator contacts. Particle colours cycle through the standard palette purely for legibility. A single mid-flow snapshot contains all three dynamic regimes simultaneously — free-fall in the reservoir, dense rearrangement and jamming in the funnel, and large-deformation outlet flow — making this a stringent stability and accuracy test bed.

B. Closing-rate stability watchdog

The integrator returns, alongside each step's new state, the maximum *closing rate* of any active contact during that step normalised by the pair's combined contact radius. For each particle-particle candidate (a, b) and each particle-wall primitive (a, s) ,

$$\rho_{\text{contact}} = \max \left\{ \max_{(a,b) \in \text{CapCand}} \frac{\|\Delta \mathbf{x}_a - \Delta \mathbf{x}_b\|}{r_{c,a} + r_{c,b}}, \max_{(a,s)} \frac{\|\Delta \mathbf{x}_a\| + |\mathbf{v}_{w,s}|}{r_{c,a} + r_{c,w,s}} \right\} \quad (54)$$

where $\Delta \mathbf{x} \equiv \mathbf{x}^{n+1} - \mathbf{x}^n$ is the per-node step displacement, $\mathbf{v}_{w,s}$ is the prescribed wall velocity for primitive s (translation, rotation arm, and oscillation amplitude $\times$ frequency, summed conservatively), and the wall radius $r_{c,w,s}$ is normally zero. The metric is *symmetric*: bulk translation that moves contacting particles together leaves $\Delta \mathbf{x}_a - \Delta \mathbf{x}_b \approx 0$ and does not trigger the watchdog. In dense DEM literature a contact should not close by more than a small fraction of r_c in a single step or contact

resolution begins to fail. Three empirical thresholds are tracked at run time:

- $\rho_{\text{contact}} \geq 0.05$: *advisory* — approaching the closing-rate limit; one-shot warning per run.
- $\rho_{\text{contact}} \geq 0.10$: *strong* — contacts likely to unwind in a few more steps; warning fires every chunk.
- $\rho_{\text{contact}} \geq 0.20$: *critical* — numerically unstable; integration will diverge.

Each warning prints the offending value and the recommended f_{dt} reduction $f_{\text{dt}}^{\text{new}} = f_{\text{dt}} \cdot (0.05/\rho_{\text{contact}})$. The watchdog never halts the run — it only reports.

The simulator exposes two equivalent entry points to the time step, kept in sync automatically: Δt directly, or the multiplier $f_{\text{dt}} = \Delta t/\Delta t_{\text{max}}$. Setting either updates the other and propagates through every TF tensor that carries the value, so users can switch between absolute and CFL-relative reasoning without rebuilding the system.

C. Matched-time drift comparison

For each f_{dt} in a sweep, the simulator runs from the 50000-step checkpoint for the same simulated time $T = 2\tau_0$, that is $N_{\text{step}} = \lceil T/\Delta t \rceil$ steps. The per-particle drift relative to the most accurate run (smallest f_{dt} in the sweep) is

$$\delta_i = \|\mathbf{x}_{\text{cm},i}^T(f_{\text{dt}}) - \mathbf{x}_{\text{cm},i}^T(f_{\text{dt}} = 0.4)\|, \quad (55)$$

and we report the per-population mean and maximum. This separates *integration error* (tracked by drift) from *instability* (tracked by ρ_{step}).

D. Sweep results across three working points

Tables XIII–XV give the sweep at three representative working points: emulsion at $\text{Bo} \in \{0.05, 0.10\}$ and elastic at $\nu = 0.5$. All use $E = 64$ (no candidate drops; see Sec. XIII E).

Findings.

- Across the entire *stable* range of f_{dt} , per-particle drift remains below 1% of R_0 even at the most aggressive settings. The integrator is integrating the same physics, only more coarsely in time — jammed and free-falling particles drift identically, with no regime-specific divergence (δ_{std} is uniformly small). This rules out the failure mode of “contact-active particles diverge first while free-fall agrees”.
- The cliff between stability and divergence is sharp. In all three configurations $\rho_{\text{contact}}^{\text{max}}$ jumps roughly an order of magnitude over a single f_{dt} step — 1.5 $\rightarrow$ 2.0 at $\text{Bo} = 0.10$ (0.025 $\rightarrow$ 0.17, 6.9 $\times$); 2.0 $\rightarrow$ 3.0 elsewhere (e.g., emulsion $\text{Bo} = 0.05$ goes 0.026 $\rightarrow$ 0.47, 18 $\times$).

TABLE XIV. Emulsion, $\text{Bo} = 0.05$, $\kappa = 0.02$, $\text{Oh} = 0.15$. Drift in units of R_0 . $\text{REF} = f_{\text{dt}} = 0.4$.

f_{dt}	$\rho_{\text{contact}}^{\text{max}}$	$\langle \delta \rangle$	δ_{max}	status
0.40	0.0022	0	0	REF
0.50	0.0027	0.0001	0.0001	clean
0.60	0.0033	0	0	clean
0.80	0.0045	0	0.0001	clean
1.00	0.0057	0.0001	0.0001	clean
1.20	0.0071	0	0.0001	clean
1.50	0.0094	0.0001	0.0002	clean
2.00	0.0256	0.0004	0.0019	clean
3.00	0.4738	0.0029	0.0509	critical

TABLE XV. Emulsion, $\text{Bo} = 0.10$ (twice the gravity / driving force). Mid-flow steady state.

f_{dt}	$\rho_{\text{contact}}^{\text{max}}$	$\langle \delta \rangle$	δ_{max}	status
0.40	0.0066	0	0	REF
0.50	0.0082	0.0002	0.0003	clean
0.60	0.0099	0	0.0001	clean
0.80	0.0132	0	0.0001	clean
1.00	0.0164	0.0002	0.0003	clean
1.20	0.0197	0	0.0003	clean
1.50	0.0247	0.0002	0.0007	clean
2.00	0.1707	0.0018	0.0167	strong
3.00	527	0.095	5.17	critical

- Stiffer modes pull the cliff in. Emulsion at $\text{Bo} = 0.05$ has plenty of headroom at $f_{\text{dt}} = 1.5$ ($\rho_{\text{contact}} = 0.009$, factor of 5 below advisory). At $\text{Bo} = 0.10$ the same f_{dt} is still clean but with only a factor of 2 of safety margin. Elastic at $\nu = 0.5$ has $\rho_{\text{contact}} \approx 0.05$ at $f_{\text{dt}} = 1.5$ — right at the advisory threshold.
- Because the watchdog covers both particle–particle and particle–wall pairs, in geometries dominated by wall contacts (such as the hopper) the maximum is set by particle–wall closing rates; in bulk emulsion or shear flow the particle–particle term dominates. No tuning is required to switch between regimes.

Recommended defaults. $f_{\text{dt}} = 1.5$ for emulsion at working-point $\kappa = 0.02$, $\text{Bo} \leq 0.05$, and $f_{\text{dt}} = 1.0$ for elastic capsules with $\nu \geq 0.5$. In a mixed simulation containing both particle types, take the minimum of the per-population recommendations; in a polydisperse sample take the minimum over sizes. Run with the watchdog enabled and let any advisory crossing be the trigger to lower f_{dt} rather than choosing it conservatively up front. A separate practical lesson from the sweeps: *transient phases* (first few thousand steps when particles are first released) are more demanding than steady-state mid-flow because of the larger initial relative velocities; we observed $f_{\text{dt}} = 1.5$ trip *critical* on the first chunk of the

TABLE XVI. Elastic, $\nu = 0.5$, $Bo = 0.05$. $\Delta t_{\max}$ is $\sim 5\times$ smaller than emulsion because the membrane mode is stiffer.

f_{dt}	$\rho_{\text{contact}}^{\max}$	$\langle \delta \rangle$	$\delta_{\max}$	status
0.40	0.0143	0	0	REF
0.50	0.0179	0.0003	0.0005	clean
0.60	0.0214	0.0003	0.0006	clean
0.80	0.0286	0.0001	0.0054	clean
1.00	0.0356	0.0004	0.0049	clean
1.20	0.0425	0.0014	0.0060	clean
1.50	0.0527	0.0005	0.0054	advisory
2.00	0.0814	0.0022	0.0104	advisory
3.00	–	–	–	NaN

$Bo = 0.10$ setup even though the subsequent steady-state was comfortably stable at the same f_{dt} .

E. Choosing the candidacy width E

The candidacy matrix has $K = P \cdot N$ rows (one per source edge) and E columns (max neighbour edges per

source). Direct measurement at the $Bo = 0.05$ checkpoint shows the matrix is heavily over-provisioned at the legacy $E = 128$ default: max active per row is 30, 99th percentile is 30, total non-zero fraction is 3.8%. Cutting E to 32 (matching the geometric peak) saves the inter-capsule force kernel a factor of ~ 4 in tensor work and is the dominant single optimisation.

The trade-off is that the geometric peak grows transiently during peak compression events. At $Bo = 0.05$ the peak rarely exceeds 32 (< 1 “row full” candidate-drop event in a 50000-step run); at $Bo = 0.10$ the peak reaches 33–34 during multiple compression events per run, producing tens of dropped candidates and visible state contamination. We recommend $E = 64$ as the safe production choice across the parameter range explored here, with $E = 32$ available for raw speed in low-Bond / low-deformation runs where the candidacy manager warning system is monitored. The candidacy manager prints two warning levels: “near capacity” (informational) at peak $\geq E - 2$, and “row full” (physics-affecting) at peak $\geq E$.

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